A similar test is available for positive definite and semidefinite matrices: The results for these matrices parallel conditions (i) to (iii) of Theorem M.D.2, but omit the factor $(-1)^r$.[9]

**Theorem M.D.3:** Let $M$ be an $N \times N$ symmetric matrix and let $B$ be an $N \times S$ matrix with $S \leq N$ and rank equal to $S$.

(i) $M$ is negative definite on $\{z \in \mathbb{R}^N: Bz = 0\}$ (i.e., $z \cdot Mz < 0$ for any $z \in \mathbb{R}^N$ with $Bz = 0$ and $z \neq 0$) if and only if

$$(-1)^r \begin{vmatrix} {}_rM_r & {}_rB \\ ({}_rB)^{\mathrm{T}} & 0 \end{vmatrix} > 0$$

for $r = S + 1, \ldots, N$.

(ii) $M$ is negative semidefinite on $\{z \in \mathbb{R}^N: Bz = 0\}$ (i.e., $z \cdot Mz \leq 0$ for any $z \in \mathbb{R}^N$ with $Bz = 0$ and $z \neq 0$) if and only if

$$(-1)^r \begin{vmatrix} {}_rM_r^\pi & {}_rB^\pi \\ ({}_rB^\pi)^{\mathrm{T}} & 0 \end{vmatrix} \geq 0$$

for $r = S + 1, \ldots, N$ *and* and every permutation $\pi$, where ${}_rB^\pi$ is the matrix formed by permuting only the *rows* of the matrix ${}_rB$ according to the permutation $\pi$ (${}_rM_r^\pi$ is, as before, a matrix formed by permuting *both* the rows *and* columns of ${}_rM_r$).

**Proof:** We will not prove this result. Note that it is parallel to parts (i) and (ii) of Theorem M.D.2 with the bordered matrix here playing a role similar to the matrix there. ■

**Example M.D.2:** Suppose we have a function of two variables, $f(x_1, x_2)$. We assume that $\nabla f(x) \neq 0$ for every $x$. Theorem M.C.4 tells us that $f(\cdot)$ is strictly quasiconcave if the Hessian matrix $D^2 f(x_1, x_2)$ is negative definite in the subspace $\{z \in \mathbb{R}^2: \nabla f(x) \cdot z = 0\}$ for every $x = (x_1, x_2)$. By Theorem M.D.3 the latter is true if and only if

$$\begin{vmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) & f_1(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) & f_2(x_1, x_2) \\ f_1(x_1, x_2) & f_2(x_1, x_2) & 0 \end{vmatrix} > 0,$$

or equivalently, if and only if

$$2f_1(x_1, x_2)f_2(x_1, x_2)f_{12}(x_1, x_2) - [f_1(x_1, x_2)]^2 f_{22}(x_1, x_2) - [f_2(x_1, x_2)]^2 f_{11}(x_1, x_2) > 0.$$

If we apply this test to $f(x_1, x_2) = x_1 x_2$ we get $2x_1 x_2 > 0$ confirming that the function is strictly quasiconcave.

By Theorem M.C.4, $f(\cdot)$ is quasiconcave if and only if the Hessian matrix $D^2 f(x_1, x_2)$ is negative semidefinite in the subspace $\{z \in \mathbb{R}^2: \nabla f(x) \cdot z = 0\}$ for every $x = (x_1, x_2)$. By Theorem M.D.3 this is true if and only if

$$\begin{vmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) & f_1(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) & f_2(x_1, x_2) \\ f_1(x_1, x_2) & f_2(x_1, x_2) & 0 \end{vmatrix} \geq 0,$$

9. Recall that $M$ is positive (semi)definite if and only if $-M$ is negative (semi)definite. Moreover, $|-{}_rM_r| = (-1)^r |{}_rM_r|$.

and (performing the appropriate permutations)

$$\begin{vmatrix} f_{22}(x_1, x_2) & f_{21}(x_1, x_2) & f_2(x_1, x_2) \\ f_{12}(x_1, x_2) & f_{11}(x_1, x_2) & f_1(x_1, x_2) \\ f_2(x_1, x_2) & f_1(x_1, x_2) & 0 \end{vmatrix} \geq 0.$$

Computing these two determinants gives us the necessary and sufficient condition

$$2f_1(x_1, x_2)f_2(x_1, x_2)f_{12}(x_1, x_2) - [f_1(x_1, x_2)]^2 f_{22}(x_1, x_2) - [f_2(x_1, x_2)]^2 f_{11}(x_1, x_2) \geq 0.$$

■

To characterize matrices that are positive definite or positive semidefinite on the subspace $\{z \in \mathbb{R}^N: \mathrm{B}z = 0\}$, we need only alter Theorem M.D.3 by replacing the term $(-1)^r$ with $(-1)^S$.

**Theorem M.D.4:** Suppose that $M$ is an $N \times N$ matrix and that for some $p \gg 0$ we have $Mp = 0$ and $M^{\mathrm{T}}p = 0$. Denote $T_p = \{z \in \mathbb{R}^N: p \cdot z = 0\}$ and let $\hat{M}$ be the $(N-1) \times (N-1)$ matrix obtained from $M$ by deleting one row and the corresponding column.

(i) If rank $M = N - 1$, then rank $\hat{M} = N - 1$.
(ii) If $z \cdot Mz < 0$ for all $z \in T_p$ with $z \neq 0$ (i.e., if $M$ is negative definite on $T_p$), then $z \cdot Mz < 0$ for any $z \in \mathbb{R}^N$ not proportional to $p$.
(iii) The matrix $M$ is negative definite on $T_p$ if and only if $\hat{M}$ is negative definite.

**Proof:** (i) Suppose that rank $\hat{M} < N - 1$, that is, $\hat{M}\hat{z} = 0$ for some $\hat{z} \in \mathbb{R}^{N-1}$ with $\hat{z} \neq 0$. Complete $\hat{z}$ to a vector $z \in \mathbb{R}^N$ by letting the value of the missing coordinate be zero. Then we have that, first, $z$ is linearly independent of $p$ (recall that $p \gg 0$) and, second, $Mz = 0$ and $Mp = 0$. Thus, rank $M < N - 1$, which contradicts the hypothesis.

(ii) Take a $z \in \mathbb{R}^N$ not proportional to $p$. For $\alpha_z = (p \cdot z)/(p \cdot p)$ and $z^* = z - \alpha_z p$, we have $z^* \in T_p$ and $z^* \neq 0$. Because $M^{\mathrm{T}}p = Mp = 0$, we have then

$$z \cdot Mz = (z^* + \alpha_z p) \cdot M(z^* + \alpha_z p) = z^* \cdot Mz^* < 0.$$

(iii) This is similar to part (ii). In fact, part (ii) directly implies that $\hat{M}$ is negative definite if $M$ is negative definite on $T_p$ (because for any $\hat{z} \in \mathbb{R}^{N-1}$, $\hat{z} \cdot \hat{M}\hat{z} = z \cdot Mz$, where $z$ has been completed from $\hat{z}$ by placing a zero in the missing coordinate, and if $\hat{z} \neq 0$ this $z$ is by construction not proportional to $p$). For the converse, let $n$ denote the row and column dropped from $M$ to obtain $\hat{M}$. If for every $z' \in T_p$ with $z' \neq 0$ we let $z = z' - (z'_n/p_n)p$, then $z_n = 0$ and $z \neq 0$ [if $z$ were equal to zero, then we would have $z' = (z'_n/p_n)p$ in contradiction to $z' \cdot p = 0$]. Moreover, $z' \cdot Mz' = z \cdot Mz = \hat{z} \cdot \hat{M}\hat{z} < 0$. ■

**Definition M.D.2:** The $N \times N$ matrix $M$ with generic entry $a_{ij}$ has a *dominant diagonal* if there is $(p_1, \ldots, p_N) \gg 0$ such that, for every $i = 1, \ldots, N$, $|p_i a_{ii}| > \sum_{j \neq i} |p_j a_{ij}|$.

**Definition M.D.3:** The $N \times N$ matrix $M$ has the *gross substitute sign pattern* if every nondiagonal entry is positive.

**Theorem M.D.5:** Suppose that $M$ is an $N \times N$ matrix.

(i) If $M$ has a dominant diagonal, then it is nonsingular.
(ii) Suppose that $M$ is symmetric. If $M$ has a negative and dominant diagonal then it is negative definite.

(iii) If $M$ has the gross substitute sign pattern and if for some $p \gg 0$ we have $Mp \ll 0$ and $M^{\mathrm{T}}p \ll 0$, then $M$ is negative definite.
(iv) If $M$ has the gross substitute sign pattern and we have $Mp = M^{\mathrm{T}}p = 0$ for some $p \gg 0$, then $\hat{M}$ is negative definite, where $\hat{M}$ is any $(N-1) \times (N-1)$ matrix obtained from $M$ by deleting a row and the corresponding column.
(v) Suppose that all the entries of $M$ are nonnegative and that $Mz \ll z$ for some $z \gg 0$ (i.e., $M$ is a *productive input–output* matrix). Then the matrix $(I - M)^{-1}$ exists. In fact, $(I - M)^{-1} = \sum_{k=0}^{k=\infty} M^k$.

**Proof:** (i) Assume, for simplicity, that $p = (1, \ldots, 1)$. Suppose, by way of contradiction, that $Mz = 0$ for $z \neq 0$. Choose a coordinate $n$ such that $|z_n| \geq |z_{n'}|$ for every other coordinate $n'$. Then $|a_{nn}z_n| > \sum_{j \neq n} |a_{nj}z_n| \geq \sum_{j \neq n} |a_{nj}z_j|$, where $a_{ij}$ is the generic entry of $M$. Hence, we cannot have $\sum_j a_{nj}z_j = 0$, and so $Mz \neq 0$. Contradiction.

(ii) If $M$ has a negative dominant diagonal then so does the matrix $M - \alpha I$, for any value $\alpha \geq 0$. Hence, by (i) we have $(-1)^N|M - \alpha I| \neq 0$. Now if $\alpha$ is very large it is clear that $(-1)^N|M - \alpha I| > 0$ (since $(-1)^N|M - \alpha I| = (-1)^N \alpha^N |(M/\alpha) - I|$ and $|-I| = (-1)^N$). Moreover, since $(-1)^N|M - \alpha I|$ is continuous in $\alpha$ and $(-1)^N|M - \alpha I| \neq 0$ for all $\alpha \geq 0$, this tells us that $(-1)^N|M - \alpha I| > 0$ for all $\alpha \geq 0$. Hence, $(-1)^N|M| > 0$. By the same argument, $(-1)^r|{}_rM_r| > 0$ for all $r$. So, if $M$ is also symmetric then by part (i) of Theorem M.D.2 it is negative definite.

(iii) The stated conditions imply that $M + M^{\mathrm{T}}$ has a negative and dominant diagonal [in particular, note that $Mp \ll 0$ and $M^{\mathrm{T}}p \ll 0$ implies that $p_n(2a_{nn}) < -\sum_{j \neq n} p_j(a_{jn} + a_{nj})$ for all $n$, where $a_{ij}$ is the generic entry of $M$]. Because, by the gross substitute property, $a_{ij} > 0$ for $i \neq j$, this gives us $|p_n(2a_{nn})| > |\sum_{j \neq n} p_j(a_{jn} + a_{nj})|$ for all $n$. Hence, the conclusion follows from part (ii) of this theorem and part (i) of Theorem M.D.1.

(iv) If $M$ satisfies the condition of (iv), then the fact that $M$ has the gross substitute sign pattern implies that $\hat{M}$ does as well and that $\hat{M}p \ll 0$ and $\hat{M}^{\mathrm{T}}p \ll 0$. Hence, $\hat{M}$ satisfies the conditions of (iii) and is therefore negative definite.

(v) This result was already proved in the Appendix to Chapter 5 (see the proof of Proposition 5.AA.1).

# M.E The Implicit Function Theorem

The setting for the *implicit function theorem* (IFT) is as follows. We have a system of $N$ equations depending on $N$ endogenous variables $x = (x_1, \ldots, x_N)$ and $M$ parameters $q = (q_1, \ldots, q_M)$:

$$\begin{gathered} f_1(x_1, \ldots, x_N; q_1, \ldots, q_M) = 0 \\ \vdots \\ f_N(x_1, \ldots, x_N; q_1, \ldots, q_M) = 0 \end{gathered} \tag{M.E.1}$$

The domain of the endogenous variables is $A \subset \mathbb{R}^N$ and the domain of the parameters is $B \subset \mathbb{R}^M$.[10]

10. In what follows, we take $A$ and $B$ to be open sets (see Section M.F) so as to avoid boundary problems.

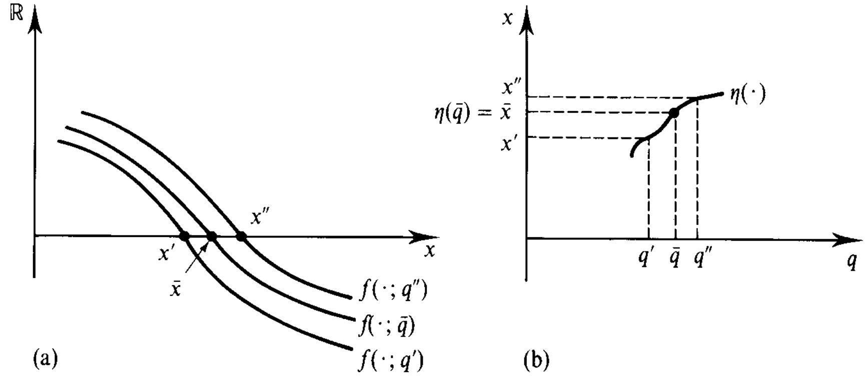


**Figure M.E.1**
A locally solvable equation.
(a) Solutions of $f(x; q) = 0$ near $(\bar{x}, \bar{q})$.
(b) The graph of $\eta(\cdot)$.

Suppose that $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_N) \in A$ and $\bar{q} = (\bar{q}_1, \ldots, \bar{q}_M) \in B$ satisfy equations (M.E.1). That is, $f_n(\bar{x}, \bar{q}) = 0$ for every $n$. We are then interested in the possibility of solving for $x = (x_1, \ldots, x_N)$ as a function of $q = (q_1, \ldots, q_M)$ *locally* around $\bar{q}$ and $\bar{x}$. Formally, we say that a set $A'$ is an *open neighborhood* of a point $x \in \mathbb{R}^N$ if $A' = \{x' \in \mathbb{R}^N \colon \|x' - x\| < \varepsilon\}$ for some scalar $\varepsilon > 0$. An open neighborhood $B'$ of a point $q \in \mathbb{R}^M$ is defined in the same way.

**Definition M.E.1:** Suppose that $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_N) \in A$ and $\bar{q} = (\bar{q}_1, \ldots, \bar{q}_M) \in B$ satisfy the equations (M.E.1). We say that we can *locally solve* equations (M.E.1) at $(\bar{x}, \bar{q})$ for $x = (x_1, \ldots, x_N)$ as a function of $q = (q_1, \ldots, q_M)$ if there are open neighborhoods $A' \subset A$ and $B' \subset B$, of $\bar{x}$ and $\bar{q}$, respectively, and $N$ uniquely determined "implicit" functions $\eta_1(\cdot), \ldots, \eta_N(\cdot)$ from $B'$ to $A'$ such that

$$f_n(\eta_1(q), \ldots, \eta_N(q); q) = 0 \qquad \text{for every } q \in B' \text{ and every } n,$$

and

$$\eta_n(\bar{q}) = \bar{x}_n \qquad \text{for every } n.$$

In Figure M.E.1 we represent, for the case where $N = M = 1$, a situation in which the system of equations can be locally solved around a given solution.

The implicit function theorem gives a sufficient condition for the existence of such implicit functions and tells us the first-order comparative statics effects of $q$ on $x$ at a solution.

**Theorem M.E.1:** (*Implicit Function Theorem*) Suppose that every equation $f_n(\cdot)$ is continuously differentiable with respect to its $N + M$ variables and that we consider a solution $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_N)$ at parameter values $\bar{q} = (\bar{q}_1, \ldots, \bar{q}_M)$, that is, satisfying $f_n(\bar{x}; \bar{q}) = 0$ for every $n$. If the *Jacobian* matrix of the system (M.E.1) with respect to the endogenous variables, evaluated at $(\bar{x}, \bar{q})$, is nonsingular, that is, if

$$\begin{vmatrix} \dfrac{\partial f_1(\bar{x}, \bar{q})}{\partial x_1} & \cdots & \dfrac{\partial f_1(\bar{x}, \bar{q})}{\partial x_N} \\ & \ddots & \\ \dfrac{\partial f_N(\bar{x}, \bar{q})}{\partial x_1} & \cdots & \dfrac{\partial f_N(\bar{x}, \bar{q})}{\partial x_N} \end{vmatrix} \neq 0, \qquad \text{(M.E.2)}$$

then the system can be locally solved at $(\bar{x}, \bar{q})$ by implicitly defined functions $\eta_n \colon B' \to A'$ that are continuously differentiable. Moreover, the first-order effects of $q$ on $x$ at $(\bar{x}, \bar{q})$ are given by

$$D_q\eta(\bar{q}) = -[D_x f(\bar{x}; \bar{q})]^{-1} D_q f(\bar{x}; \bar{q}). \qquad \text{(M.E.3)}$$

**Proof:** A proof of the existence of the implicit functions $\eta_n$: $B' \to A'$ is too technical for this appendix, but its common-sense logic is easy to grasp. Expression (M.E.2), a full rank condition, tells us that we can move the values of the system of equations in any direction by appropriate changes of the endogenous variables. Therefore, if there is a shock to the parameters and the values of the equation system are pushed away from zero, then we can adjust the endogenous variables so as to restore the "equilibrium."

Now, given a system of implicit functions $\eta(q) = (\eta_1(q), \ldots, \eta_N(q))$ defined on some neighborhood of $(\bar{x}, \bar{q})$, the first-order comparative static effects $\partial\eta_n(\bar{q})/\partial q_m$ are readily determined. Let $f(x; q) = (f_1(x; q), \ldots, f_N(x; q))$. Since we have

$$f(\eta(q); q) = 0 \qquad \text{for all } q \in B',$$

we can apply the chain rule of calculus to obtain

$$D_x f(\bar{x}; \bar{q}) D_q \eta(\bar{q}) + D_q f(\bar{x}; \bar{q}) = 0.$$

Because of (M.E.2), the $N \times N$ matrix $D_x f(\bar{x}; \bar{q})$ is invertible, and so we can conclude that

$$D_q\eta(\bar{q}) = -[D_x f(\bar{x}; \bar{q})]^{-1} D_q f(\bar{x}; \bar{q}). \qquad \blacksquare$$

Note that when $N = M = 1$ (the case of one endogenous variable and one parameter), (M.E.3) reduces to the simple expression

$$\frac{d\eta(\bar{q})}{dq} = -\frac{\partial f(\bar{x}; \bar{q})/\partial q}{\partial f(\bar{x}; \bar{q})/\partial x}.$$

The special case of the implicit function theorem where $M = N$ and every equation has the form $f_n(x, q) = g_n(x) - q_n = 0$ is known as the *inverse function theorem.*

---

How restrictive is condition (M.E.2)? Not very. In Figure M.E.2 we depict a situation where it fails to hold. [By contrast, in Figure M.E.1 condition (M.E.2) is satisfied.] However, the tangency displayed in Figure M.E.2 appears pathological: it would be removed by any small perturbation of the function $f(\cdot\,; \cdot)$.

An important result, the *transversality theorem,* makes this idea precise by asserting that, under a weak condition [enough first-order variability of $f(\cdot\,; \cdot)$ with respect to $x$ *and* $q$], (M.E.2) holds *generically* on the parameters. We present a preliminary concept in Definition M.E.2.

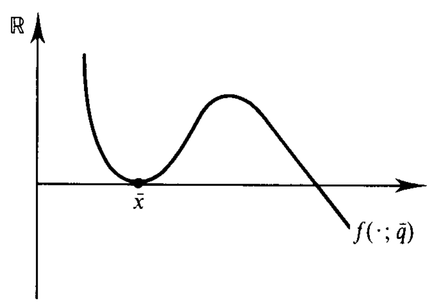


**Figure M.E.2**
Condition (M.E.2) is violated at the solution $(\bar{x}, \bar{q})$.

**Definition M.E.2:** Given open sets $A \subset \mathbb{R}^N$ and $B \subset \mathbb{R}^M$, the (continuously differentiable) system of equations $f(\cdot\,; \hat{q}) = 0$ defined on $A$ is *regular at* $\hat{q} \in B$ if (M.E.2) holds at any solution $x$; that is, if $f(x; \hat{q}) = 0$ implies that $|D_x f(x; \hat{q})| \neq 0$.

With this definition we then have Theorem M.E.2.

**Theorem M.E.2:** (*Transversality Theorem*) Suppose that we are given open sets $A \subset \mathbb{R}^N$ and $B \subset \mathbb{R}^M$ and a (continuously differentiable) function $f: A \times B \to \mathbb{R}^N$. If $f(\cdot\,; \cdot)$ satisfies the condition

The $N \times (N + M)$ matrix $Df(x; q)$ has rank $N$ whenever $f(x; q) = 0$,

then the system of $N$ equations in $N$ unknowns $f(\cdot\,; \hat{q}) = 0$ is regular for *almost every* $\hat{q} \in B$.[11]

# M.F Continuous Functions and Compact Sets

In this section, we begin by formally defining the concept of a continuous function. We then develop the notion of a compact set (and, along the way, the notions of open and closed sets). Finally, we discuss some properties of continuous functions that relate to compact sets.

A *sequence* in $\mathbb{R}^N$ assigns to every positive integer $m = 1, 2, \ldots$ a vector $x^m \in \mathbb{R}^N$. We denote the sequence by $\{x^m\}_{m=1}^{m=\infty}$ or, simply, by $\{x^m\}$ or even $x^m$.

**Definition M.F.1:** The sequence $\{x^m\}$ *converges* to $x \in \mathbb{R}^N$, written as $\lim_{m \to \infty} x^m = x$, or $x^m \to x$, if for every $\varepsilon > 0$ there is an integer $M_\varepsilon$ such that $\|x^m - x\| < \varepsilon$ whenever $m > M_\varepsilon$. The point $x$ is then said to be the *limit point* (or simply the *limit*) of sequence $\{x^m\}$.

In words: The sequence $\{x^m\}$ converges to $x$ if $x^m$ approaches $x$ arbitrarily closely as $m$ increases.

**Definition M.F.2:** Consider a domain $X \subset \mathbb{R}^N$. A function $f: X \to \mathbb{R}$ is *continuous* if for all $x \in X$ and every sequence $x^m \to x$ (having $x^m \in X$ for all $m$), we have $f(x^m) \to f(x)$. A function $f: X \to \mathbb{R}^K$ is continuous if every coordinate function $f_k(\cdot)$ is continuous.

In words: a function is continuous if, when we take a sequence of points $x^1, x^2, \ldots$ converging to $x$, the corresponding sequence of function values $f(x^1), f(x^2), \ldots$ converges to $f(x)$. Intuitively, a function fails to be continuous if it displays a "jump" in its value at some point $x$. Examples of continuous and discontinuous functions defined on $[0, 1]$ are illustrated in Figure M.F.1.

We next develop the notions of open, closed, and compact sets.

**Definition M.F.3:** Fix a set $X \subset \mathbb{R}^N$. We say that a set $A \subset X$ is *open* (relative to $X$) if for every $x \in A$ there is an $\varepsilon > 0$ such that $\|x' - x\| < \varepsilon$ and $x' \in X$ implies $x' \in A$.

11. "Almost every" means that if, for example, we choose $\hat{q}$ according to some nondegenerate multinomial normal distribution in $\mathbb{R}^M$, then with probability 1 the equation system $f(\cdot\,; \hat{q}) = 0$ is regular. *This is the concept of "genericity" in this context.*

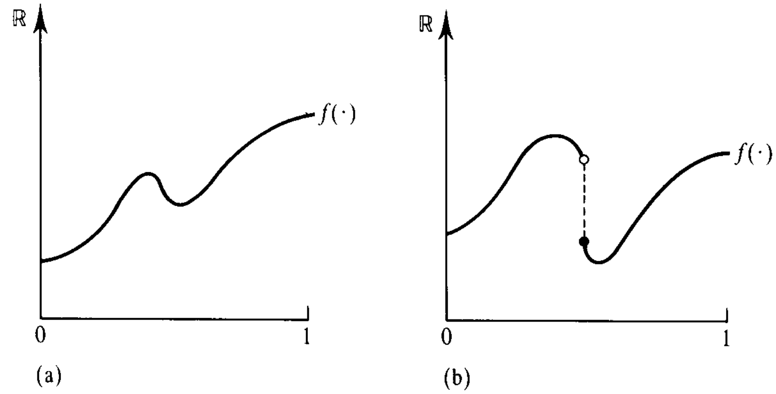


**Figure M.F.1**
Continuous and discontinuous functions.
(a) A continuous function.
(b) A discontinuous function.

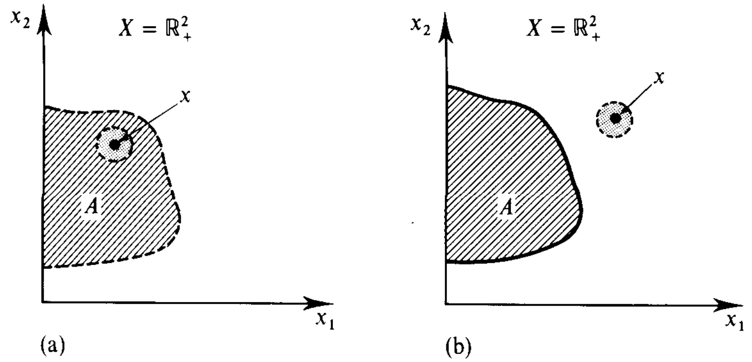


**Figure M.F.2**
Open and closed sets.
(a) An open set (relative to $X$).
(b) A closed set (relative to $X$).

A set $A \subset X$ is *closed* (relative to $X$) if its complement $X \backslash A$ is open (relative to $X$).[12] If $X = \mathbb{R}^N$ we simply refer to "open" and "closed" sets.

Given a point $x \in \mathbb{R}^N$, a set $B = \{x' \in \mathbb{R}^N \colon \|x' - x\| < \varepsilon\}$ for some scalar $\varepsilon > 0$ is called an *open ball around* $x$. With this notion, the idea of an open set can be put as follows: Suppose that the universe of possible vectors in $\mathbb{R}^N$ is $X$. A set $A \subset X$ is open (relative to $X$) if, for every point $x$ in $A$, there is an open ball around $x$ all of whose elements (in $X$) are elements of $A$. In Figure M.F.2(a) the hatched set $A$ is open (relative to $X$). In the figure, we depict a typical point $x \in A$ and a shaded open ball around $x$ that lies within $A$; points on the dashed curve do not belong to $A$. In contrast, the hatched set $A$ in Figure M.F.2(b) is closed because the set $X \backslash A$ is open; note how there is an open ball around the point $x \in X \backslash A$ that lies entirely within $X \backslash A$ [in the figure, the points on the inner solid curve belong to $A$].

Theorem M.F.1 gathers some basic facts about open and closed sets.

**Theorem M.F.1:** Fix a set $X \subset \mathbb{R}^N$. In what follows, all the open and closed sets are relative to $X$.

(i) The union of any number, finite or infinite, of open sets is open. The intersection of a finite number of open sets is open.

(ii) The intersection of any number, finite or infinite, of closed sets is closed. The union of a finite number of closed sets is closed.

(iii) A set $A \subset X$ is closed if and only if for every sequence $x^m \to x \in X$, with $x^m \in A$ for all $m$, we have $x \in A$.

12. Given two sets $A$ and $B$, the set $A \backslash B$ is the set containing all the elements of $A$ that are not elements of $B$.

Property (iii) of Theorem M.F.1 is noteworthy because it gives us a direct way to characterize a closed set: a set $A$ is closed if and only if the limit point of any sequence whose members are all elements of $A$ is itself an element of $A$. Points (in $X$) that are the limits of sequences whose members are all elements of the set $A$ are known as the *limit points* of $A$. Thus, property (iii) says that a set $A$ is closed if and only if it contains all of its limit points.

Given $A \subset X$, the *interior* of $A$ (relative to $X$) is the open set[13]

$$\text{Int}_X\, A = \{x \in A: \text{there is } \varepsilon > 0 \text{ such that } \|x' - x\| < \varepsilon \text{ and } x' \in X \text{ implies } x' \in A\}.$$

The *closure of* $A$ (relative to $X$) is the closed set $\text{Cl}_X\, A = X \backslash \text{Int}_X\, (X \backslash A)$. Equivalently, $\text{Cl}_X\, A$ is the union of the set $A$ and its limit points; it is the smallest closed set containing $A$. The *boundary* of $A$ (relative to $X$) is the closed set $\text{Bdry}_X\, A = \text{Cl}_X\, A \backslash \text{Int}_X\, A$. The set $A$ is closed if and only if $\text{Bdry}_X\, A \subset A$.

**Definition M.F.4:** A set $A \subset \mathbb{R}^N$ is *bounded* if there is $r \in \mathbb{R}$ such that $\|x\| < r$ for every $x \in A$. The set $A \subset \mathbb{R}^N$ is *compact* if it is bounded and closed relative to $\mathbb{R}^N$.

We conclude by noting two properties of continuous functions relating to compact sets. Formally, given a function $f: X \to \mathbb{R}^K$, the *image* of a set $A \subset X$ under $f(\cdot)$ is the set $f(A) = \{y \in \mathbb{R}^K: y = f(x) \text{ for some } x \in A\}$.

**Theorem M.F.2:** Suppose that $f: X \to \mathbb{R}^K$ is a continuous function defined on a nonempty set $X \subset \mathbb{R}^N$.

(i) The image of a compact set under $f(\cdot)$ is compact: That is, if $A \subset X$ is compact, then $f(A) = \{y \in \mathbb{R}^K: y = f(x) \text{ for some } x \in A\}$ is a compact subset of $\mathbb{R}^K$.

(ii) Suppose that $K = 1$ and $X$ is compact. Then $f(\cdot)$ has a maximizer: That is, there is $x \in X$ such that $f(x) \geq f(x')$ for every $x' \in X$.

Part (ii) of Theorem M.F.2 asserts that any continuous function $f: X \to \mathbb{R}$ defined on a compact set $X$ attains a maximum. We illustrate this result in Figure M.F.3. A maximum is not attained either in Figure M.F.3(a) or in Figure M.F.3(b). In Figure M.F.3(a), the function is continuous, but the domain is not compact. In Figure M.F.3(b), the domain is compact, but the function is not continuous.

---

Given a sequence $\{x^m\}$, suppose that we have a strictly increasing function $m(k)$ that assigns to each positive integer $k$ a positive integer $m(k)$. Then the sequence $x^{m(1)}, x^{m(2)}, \ldots$ (written $\{x^{m(k)}\}$) is called a *subsequence* of $\{x^m\}$. That is, $\{x^{m(k)}\}$ is composed of an (order-preserving) subset of the sequence $\{x^m\}$. For example, if the sequence $\{x^m\}$ is $1, 2, 4, 16, 25, 36, \ldots$, then one subsequence of $\{x^m\}$ is $1, 4, 16, 36, \ldots$; another is $2, 4, 16, 25, 36, \ldots$.

**Theorem M.F.3:** Suppose that the set $A \subset \mathbb{R}^N$ is compact.

(i) Every sequence $\{x^m\}$ with $x^m \in A$ for all $m$ has a convergent subsequence. Specifically, there is a subsequence $\{x^{m(k)}\}$ of the sequence $\{x^m\}$ that has a limit in $A$, that is, a point $x \in A$ such that $x^{m(k)} \to x$.

(ii) If, in addition to being compact, $A$ is also *discrete*, that is, if all its points are isolated [formally, for every $x \in A$ there is $\varepsilon > 0$ such that $x' = x$ whenever $x' \in A$ and $\|x' - x\| < \varepsilon$], then $A$ is finite.

---

13. In what follows in this paragraph, all of the open and closed sets are once again relative to $X$.

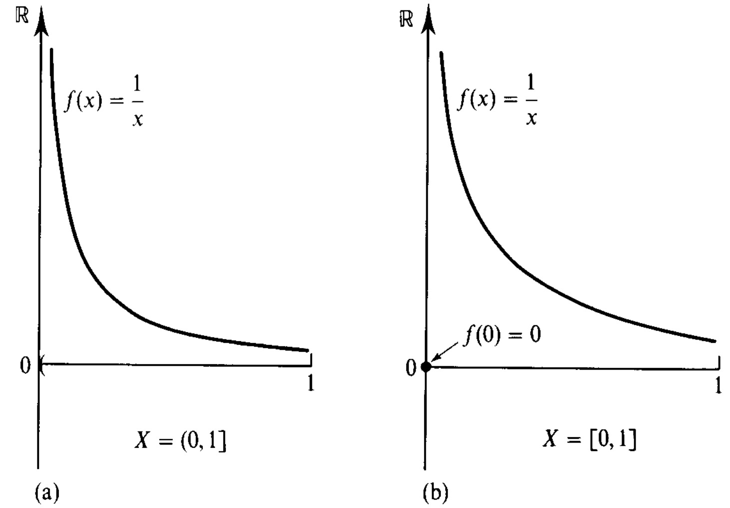


**Figure M.F.3**
Indispensability of the continuity and compactness assumptions for the existence of a maximizer.
(a) A continuous function with no maximizer on a noncompact domain.
(b) A discontinuous function with no maximizer on a compact domain.

# M.G Convex Sets and Separating Hyperplanes

In this section, we review some basic properties of convex sets, including the important separating hyperplane theorems.

**Definition M.G.1:** The set $A \subset \mathbb{R}^N$ is *convex* if $\alpha x + (1 - \alpha)x' \in A$ whenever $x, x' \in A$ and $\alpha \in [0, 1]$.[14]

In words: A set in $\mathbb{R}^N$ is convex if whenever it contains two vectors $x$ and $x'$, it also contains the entire segment connecting them. In Figure M.G.1(a), we depict a convex set. The set in Figure M.G.1(b) is not convex.

Note that for a concave function $f\colon A \to \mathbb{R}$ the set $\{(z, v) \in \mathbb{R}^{N+1}\colon v \leq f(z), z \in A\}$ is convex. Note also that the intersection of any number of convex sets is convex, but the union of convex sets need not be convex.

**Definition M.G.2:** Given a set $B \subset \mathbb{R}^N$, the *convex hull* of $B$, denoted Co $B$, is the smallest convex set containing $B$, that is, the intersection of all convex sets that contain $B$.

Figure M.G.2 represents a set and its convex hull. It is not difficult to verify that the convex hull can also be described as the set of all possible convex combinations of elements of $B$, that is,

$$\text{Co } B = \left\{ \sum_{j=1}^{J} \alpha_j x_j\colon \text{for some } x_1, \ldots, x_J \text{ with } x_j \in B \text{ for all } j, \right.$$
$$\left. \text{and some } (\alpha_1, \ldots, \alpha_J) \geq 0 \text{ with } \sum_{j=1}^{J} \alpha_j = 1 \right\}.$$

**Definition M.G.3:** The vector $x \in B$ is an *extreme point* of the convex set $B \subset \mathbb{R}^N$ if it cannot be expressed as $x = \alpha y + (1 - \alpha)z$ for any $y, z \in B$ and $\alpha \in (0, 1)$.

14. The set $A$ is *strictly convex* if $\alpha x + (1 - \alpha)x'$ is an element of the *interior* of $A$ whenever $x, x' \in A$ and $\alpha \in (0, 1)$ (see Section M.F for a definition of the interior of a set).

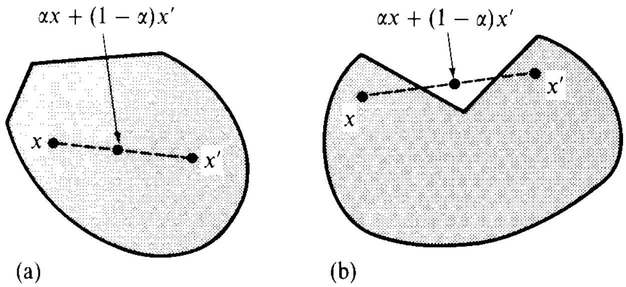


**Figure M.G.1**
Convex and nonconvex sets.
*(a) A convex set.*
(b) A nonconvex set.

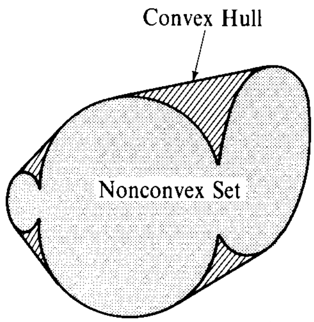


**Figure M.G.2**
A nonconvex set and its convex hull.

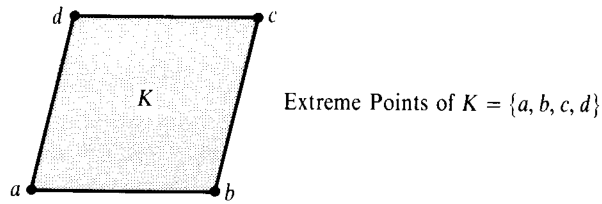


**Figure M.G.3**
Extreme points of a convex set.

The extreme points of the convex set represented in Figure M.G.3 are the four corners.

A very important result of convexity theory is contained in Theorem M.G.1.

**Theorem M.G.1:** Suppose that $B \subset \mathbb{R}^N$ is a convex set that is also compact (that is, closed and bounded; see Section M.F). Then every $x \in B$ can be expressed as a convex combination of at most $N + 1$ extreme points of $B$.

**Proof:** The proof is too technical to be given here. Note simply that the result is correct for the convex set in Figure M.G.3: Any point belongs to the triangle spanned by *some* collection of three corners. ■

We now turn to the development of the separating hyperplane theorems.

**Definition M.G.4:** Given $p \in \mathbb{R}^N$ with $p \neq 0$, and $c \in \mathbb{R}$, the *hyperplane generated by p and c* is the set $H_{p,c} = \{z \in \mathbb{R}^N: p \cdot z = c\}$. The sets $\{z \in \mathbb{R}^N: p \cdot z \geq c\}$ and $\{z \in \mathbb{R}^N: p \cdot z \leq c\}$ are called, respectively, the *half-space above* and the *half-space below* the hyperplane $H_{p,c}$.

Hyperplanes and half-spaces are convex sets. Figure M.G.4 provides illustrations.

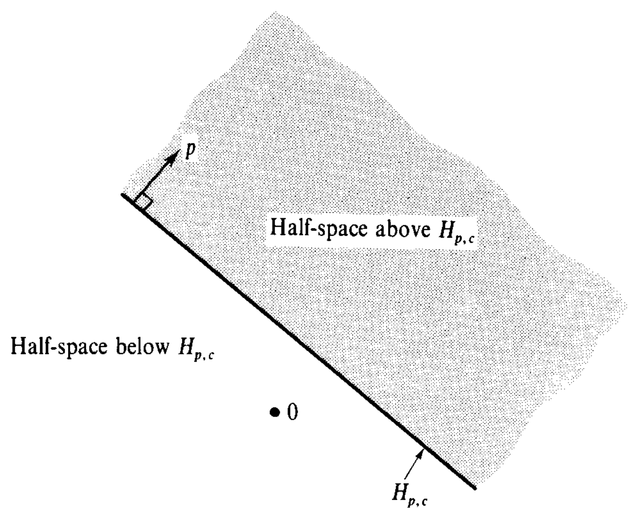


**Figure M.G.4**
Hyperplanes and half-spaces.

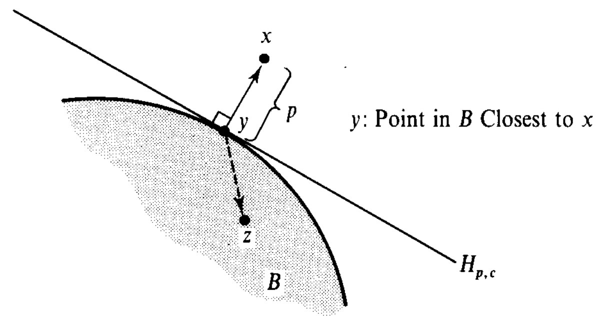


**Figure M.G.5**
The separating hyperplane theorem.

**Theorem M.G.2:** (*Separating Hyperplane Theorem*) Suppose that $B \subset \mathbb{R}^N$ is convex and closed (see Section M.F for a discussion of closed sets), and that $x \notin B$. Then there is $p \in \mathbb{R}^N$ with $p \neq 0$, and a value $c \in \mathbb{R}$ such that $p \cdot x > c$ and $p \cdot y < c$ for every $y \in B$.

More generally, suppose that the convex sets $A, B \subset \mathbb{R}^N$ are disjoint (i.e., $A \cap B = \varnothing$). Then there is $p \in \mathbb{R}^N$ with $p \neq 0$, and a value $c \in \mathbb{R}$, such that $p \cdot x \geq c$ for every $x \in A$ and $p \cdot y \leq c$ for every $y \in B$. That is, there is a hyperplane that separates $A$ and $B$, leaving $A$ and $B$ on different sides of it.

**Proof:** We discuss only the first part (i.e., the separation of a point and a closed, convex set). In Figure M.G.5 we represent a closed, convex set $B$ and a point $x \notin B$. We also indicate by $y \in B$ the point in set $B$ closest to $x$.[15] If we let $p = x - y$ and $c' = p \cdot y$, we can then see, first, that $p \cdot x > c'$ [since $p \cdot x - c' = p \cdot x - p \cdot y = (x - y) \cdot (x - y) = \|x - y\|^2 > 0$] and, second, that for any $z \in B$ the vectors $p$ and $z - y$ cannot make an acute angle, that is, $p \cdot (z - y) = p \cdot z - c' \leq 0$. Finally, let $c = c' + \varepsilon$ where $\varepsilon > 0$ is small enough for $p \cdot x > c' + \varepsilon = c$ to hold. ■

15. We use the familiar Euclidean distance measure. It is to guarantee the existence of a closest point in $B$ that we require $B$ to be closed.

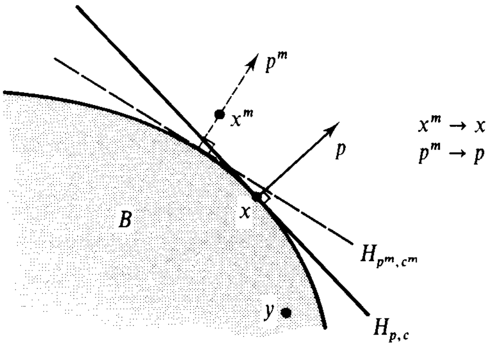


**Figure M.G.6**
The supporting hyperplane theorem

**Theorem M.G.3:** (*Supporting Hyperplane Theorem*) Suppose that $B \subset \mathbb{R}^N$ is convex and that $x$ is not an element of the interior of set $B$ (i.e., $x \notin \operatorname{Int} B$; see Section M.F for the concept of the interior of a set). Then there is $p \in \mathbb{R}^N$ with $p \neq 0$ such that $p \cdot x \geq p \cdot y$ for every $y \in B$.

**Proof:** Suppose that $x \notin \operatorname{Int} B$. The following argument can be followed in Figure M.G.6. It is intuitive that we can find a sequence $x^m \to x$ such that, for all $m$, $x^m$ is not an element of the closure of set $B$ (i.e., $x^m \notin \operatorname{Cl} B$; see Section M.F for a discussion of sequences and the closure of a set). By the separating hyperplane theorem (Theorem M.G.2), for each $m$ there is a $p^m \neq 0$ and a $c^m \in \mathbb{R}$ such that

$$p^m \cdot x^m > c^m \geq p^m \cdot y \tag{M.G.1}$$

for every $y \in B$. Without loss of generality we can suppose that $\|p^m\| = 1$ for every $m$. Thus, extracting a subsequence if necessary (see the small type discussion at the end of Section M.F), we can assume that there is $p \neq 0$ and $c \in \mathbb{R}$ such that $p^m \to p$ and $c^m \to c$. Hence, taking limits in (M.G.1), we have

$$p \cdot x \geq c \geq p \cdot y$$

for every $y \in B$. ■

Finally, for the important concept of the *support function* of a set and its properties we refer to Section 3.F of the text.

# M.H Correspondences

It is common in economics to resort to a generalized concept of a function called a *correspondence*.

**Definition M.H.1:** Given a set $A \subset \mathbb{R}^N$, a *correspondence* $f: A \to \mathbb{R}^K$ is a rule that assigns a set $f(x) \subset \mathbb{R}^K$ to every $x \in A$.

Note that when, for every $x \in A$, $f(x)$ is composed of precisely one element, then $f(\cdot)$ can be viewed as a function in the usual sense. Note also that the definition allows for $f(x) = \varnothing$, but typically we consider only correspondences with $f(x) \neq \varnothing$ for every $x \in A$. Finally, if for some set $Y \subset \mathbb{R}^K$ we have $f(x) \subset Y$ for every $x \in A$, we indicate this by $f: A \to Y$.

We now proceed to discuss continuity notions for correspondences. Given $A \subset \mathbb{R}^N$ and $Y \subset \mathbb{R}^K$, the *graph* of the correspondence $f: A \to Y$ is the set $\{(x, y) \in A \times Y: y \in f(x)\}$.

**Definition M.H.2:** Given $A \subset \mathbb{R}^N$ and the closed set $Y \subset \mathbb{R}^K$, the correspondence $f: A \to Y$ has a *closed graph* if for any two sequences $x^m \to x \in A$ and $y^m \to y$, with $x^m \in A$ and $y^m \in f(x^m)$ for every $m$, we have $y \in f(x)$.

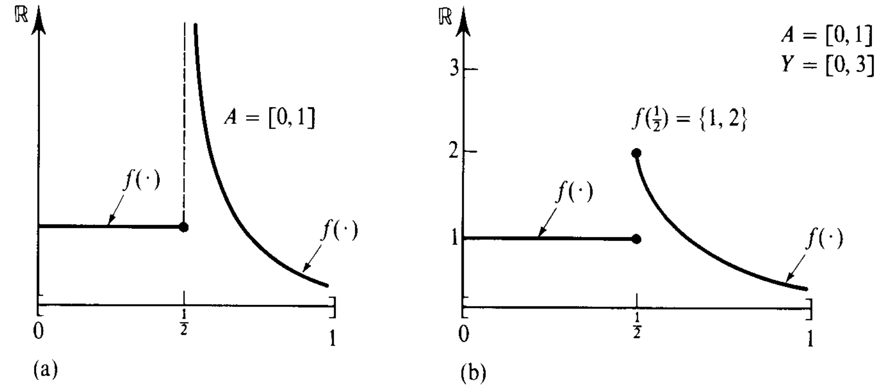


**Figure M.H.1** Closed graphs and upper hemicontinuous correspondences. (a) A closed graph correspondence that is not upper hemicontinuous. (b) An upper hemicontinuous correspondence.

Note that the concept of a closed graph is simply our usual notion of closedness (relative to $A \times Y$) applied to the set $\{(x, y) \in A \times Y: y \in f(x)\}$ (see Section M.F).

**Definition M.H.3:** Given $A \subset \mathbb{R}^N$ and the closed set $Y \subset \mathbb{R}^K$, the correspondence $f: A \to Y$ is *upper hemicontinuous* (uhc) if it has a closed graph and the images of compact sets are bounded, that is, for every compact set $B \subset A$ the set $f(B) = \{y \in Y: y \in f(x) \text{ for some } x \in B\}$ is bounded.[16,17]

In many applications, the range space $Y$ of $f(\cdot)$ is itself compact. In that case, the upper hemicontinuity property reduces to the closed graph condition. In Figure M.H.1(a), we represent a correspondence (in fact, a function) having a closed graph that is not upper hemicontinuous. In contrast, the correspondence represented in Figure M.H.1(b) is upper hemicontinuous.

The upper hemicontinuity property for correspondences can be thought of as a natural generalization of the notion of continuity for functions. Indeed, we have the result of Theorem M.H.1.

**Theorem M.H.1:** Given $A \subset \mathbb{R}^N$ and the closed set $Y \subset \mathbb{R}^K$, suppose that $f: A \to Y$ is a single-valued correspondence (so that it is, in fact, a function). Then $f(\cdot)$ is an upper hemicontinuous correspondence if and only if it is continuous as a function.

---

**Proof:** If $f(\cdot)$ is continuous as a function, then Definition M.F.2 implies that $f(\cdot)$ has a closed graph (relative to $A \times Y$). In addition, Theorem M.F.2 tells us that images of compact sets under $f(\cdot)$ are compact, hence bounded. Thus, $f(\cdot)$ is upper hemicontinuous as a correspondence.

Suppose now that $f(\cdot)$ is upper hemicontinuous as a correspondence and consider any sequence $x^m \to x \in A$ with $x^m \in A$ for all $m$. Let $S = \{x^m: m = 1, 2, \ldots\} \cup \{x\}$. Then there exists an $r > 0$ such that $\|x'\| < r$ if $x' \in S$.[18] Because $S$ is also closed, it follows that $S$ is compact.

---

16. See Section M.F for a discussion of bounded and compact sets.

17. It can be verified that Definition M.H.3 implies that the image of a compact set under an upper hemicontinuous correspondence is in fact compact (i.e., closed and bounded), a property also shared by continuous functions (see Theorem M.F.2).

18. To see this, recall that if $x^m \to x$, then for any $\varepsilon > 0$ there is a positive integer $M_\varepsilon$ such that $\|x^m - x\| < \varepsilon$ for all $m > M_\varepsilon$. Hence, for any $r > \text{Max}\,\{\|x^1\|, \ldots, \|x^{M_\varepsilon}\|, \|x\| + \varepsilon\}$, we have $\|x'\| < r$ if $x' \in S$.

By Definition M.H.3, $f(S)$ is bounded, and so $\mathrm{Cl}\, f(S)$ (relative to $\mathbb{R}^K$) is a compact set. If, contradicting continuity of the function $f(\cdot)$, the sequence $\{f(x^m)\}$ [which lies in the compact set $\mathrm{Cl}\, f(S)$] did not converge to $f(x)$, then by Theorem M.F.3 we could extract a subsequence $x^{m(k)} \to x$ such that $f(x^{m(k)}) \to y$ for some $y \in \mathrm{Cl}\, f(S)$ having $y \neq f(x)$. But then the graph of $f(\cdot)$ could not be closed, in contradiction to the upper hemicontinuity of $f(\cdot)$ as a correspondence. ■

---

Upper hemicontinuity is only one of two possible generalizations of the continuity notion to correspondences. We now state the second (for the case where the range space $Y$ is compact).

**Definition M.H.4:** Given $A \subset \mathbb{R}^N$ and a compact set $Y \subset \mathbb{R}^K$, the correspondence $f: A \to Y$ is *lower hemicontinuous* (lhc) if for every sequence $x^m \to x \in A$ with $x^m \in A$ for all $m$, and every $y \in f(x)$, we can find a sequence $y^m \to y$ and an integer $M$ such that $y^m \in f(x^m)$ for $m > M$.

Figure M.H.2(a) represents a lower hemicontinuous correspondence.[19] Observe that the correspondence is not upper hemicontinuous—it does not have a closed graph. Similarly, the correspondence represented in Figure M.H.2(b) is upper hemicontinuous but it fails to be lower hemicontinuous (consider the illustrated sequence $x^m \to x$ that approaches $x$ from below and the point $y \in f(x)$). Roughly speaking, upper hemicontinuity is compatible only with "discontinuities" that appear as "explosions" of sets [as at $x = \frac{1}{2}$ in Figure M.H.2(b)], while lower hemicontinuity is compatible only with "implosions" of sets [as at $x = \frac{1}{2}$ in Figure M.H.2(a)].

As with upper hemicontinuous correspondences, if $f(\cdot)$ is a function then the concepts of lower hemicontinuity as a correspondence and of continuity as a function coincide.

Finally, when a correspondence is both upper and lower hemicontinuous, we say that it is *continuous*. An example is illustrated in Figure M.H.3.

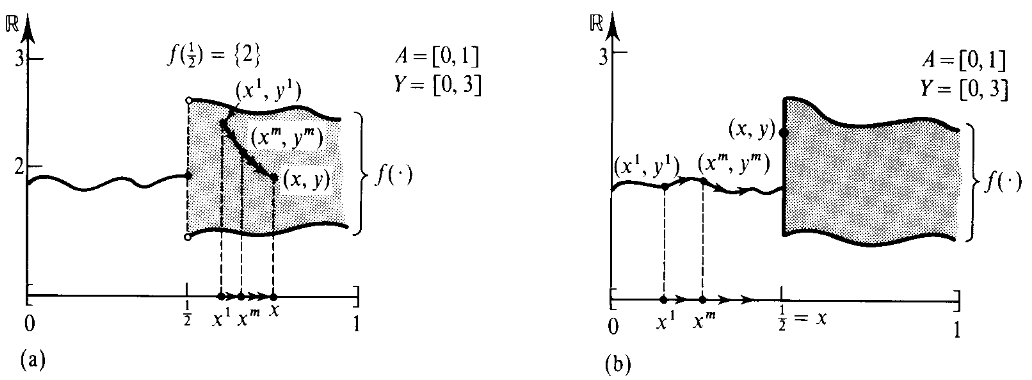


**Figure M.H.2**
Upper and lower hemicontinuous correspondences. (a) A lower hemicontinuous correspondence that is not upper hemicontinuous. (b) An upper hemicontinuous correspondence that is not lower hemicontinuous.

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19. For another source of examples, note that any correspondence $f: A \to Y$ with an open graph (relative to $A \times Y$) is lower hemicontinuous.

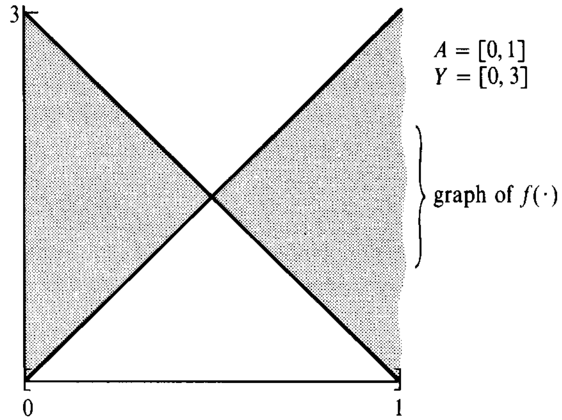


**Figure M.H.3**
A continuous correspondence.

# M.I Fixed Point Theorems

In economics the most frequent technique for establishing the existence of solutions to an equilibrium system of equations consists of setting up the problem as the search for a *fixed point* of a suitably constructed function or correspondence $f: A \to A$ from some set $A \subset \mathbb{R}^N$ into itself. A vector $x \in A$ is a fixed point of $f(\cdot)$ if $x = f(x)$ [or, in the correspondence case, if $x \in f(x)$]. That is, the vector is mapped into itself and so it remains "fixed." The reason for proceeding in this, often roundabout, way is that important mathematical theorems for proving the existence of fixed points are readily available.

The most basic and well-known result is stated in Theorem M.I.1.

**Theorem M.I.1:** (*Brouwer's Fixed Point Theorem*) Suppose that $A \subset \mathbb{R}^N$ is a nonempty, compact, convex set, and that $f: A \to A$ is a continuous function from $A$ into itself. Then $f(\cdot)$ has a fixed point; that is, there is an $x \in A$ such that $x = f(x)$.

The logic of Brouwer's fixed point theorem is illustrated in Figure M.I.1(a) for the easy case where $N = 1$ and $A = [0, 1]$. In this case, the theorem says that the graph of any continuous function from the interval $[0, 1]$ into itself must cross the diagonal, and it is then a simple consequence of the *intermediate value theorem*. In

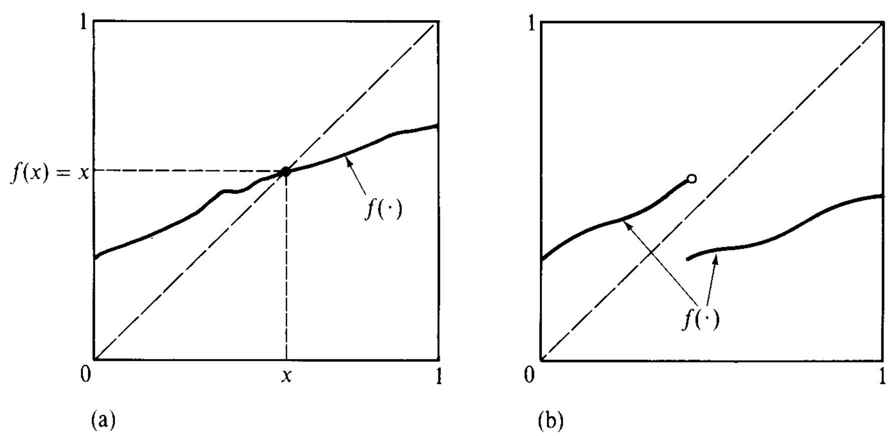


**Figure M.I.1**
Brouwer's fixed point theorem.
(a) A continuous function from [0, 1] to [0, 1] has a fixed point.
(b) The continuity assumption is indispensable.

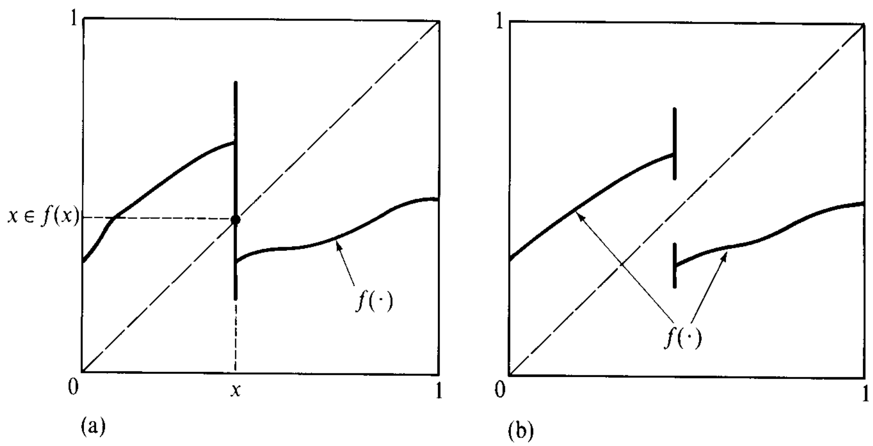


**Figure M.I.2**
Kakutani's fixed point theorem.
(a) A fixed point exists.
(b) The convex-valuedness assumption is indispensable.

particular, if we define the continuous function $\phi(x) = f(x) - x$, then $\phi(0) \geq 0$ and $\phi(1) \leq 0$, and so $\phi(x) = 0$ for some $x \in [0, 1]$; hence, $f(x) = x$ for some $x \in [0, 1]$. In Figure M.I.1(b) we can see that, indeed, the continuity of $f(\cdot)$ is required. As for the convexity of the domain, consider the function defined by a 90-degree clockwise rotation on the circle $S = \{x \in \mathbb{R}^2\colon \|x\| = 1\}$: It is a continuous function with no fixed point. The set $S$, however, is not convex.

In applications, it is often the case that the following extension of Brouwer's fixed point theorem to correspondence is most useful.

**Theorem M.I.2:** (*Kakutani's Fixed Point Theorem*) Suppose that $A \subset \mathbb{R}^N$ is a nonempty, compact, convex set, and that $f\colon A \rightarrow A$ is an upper hemicontinuous correspondence from $A$ into itself with the property that the set $f(x) \subset A$ is nonempty and convex for every $x \in A$. Then $f(\cdot)$ has a fixed point; that is, there is an $x \in A$ such that $x \in f(x)$.

The logic of Kakutani's fixed point theorem is illustrated in Figure M.I.2(a) for $N = 1$. Note that the convexity of the set $f(x)$ for all $x$ is indispensable. Without this condition we could have cases such as that in Figure M.I.2(b) where no fixed point exists.

Finally, we mention a fixed point theorem that is of a different style but that is being found of increasing relevance to economic applications.

**Theorem M.I.3:** (*Tarsky's Fixed Point Theorem*) Suppose that $f\colon [0, 1]^N \rightarrow [0, 1]^N$ is a nondecreasing function, that is, $f(x') \geq f(x)$ whenever $x' \geq x$. Then $f(\cdot)$ has a fixed point; that is, there is an $x \in A$ such that $x = f(x)$.

Tarsky's theorem differs from Brouwer's in three respects. First, the base set is not any compact, convex set, but rather a special one—an $N$-product of intervals. Second, the function is required to be nondecreasing. Third, the function is not required to be continuous. The logic of Tarsky's fixed point theorem is illustrated in Figure M.I.3 for the case $N = 1$. In the figure, the function $f(\cdot)$ is not continuous. Yet, the fact that it is nondecreasing forces its graph to intersect the diagonal.

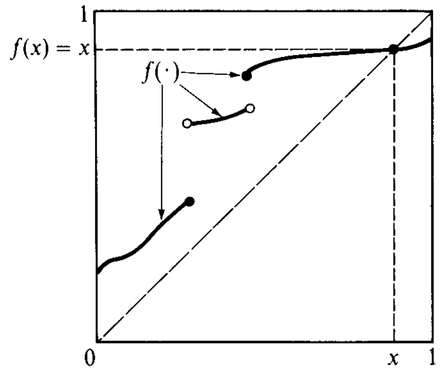


**Figure M.I.3 (left)**
Tarski's fixed point theorem.

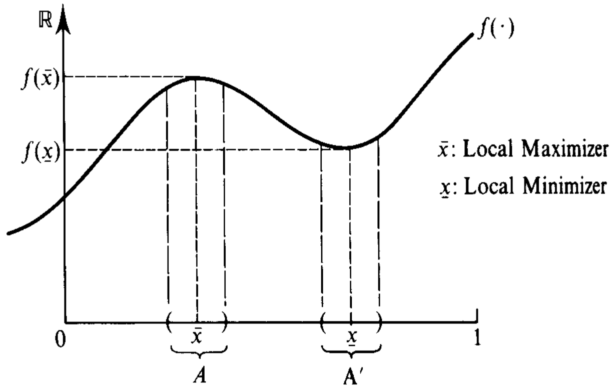


**Figure M.J.1 (right)**
Local maximizers and minimizers.

# M.J Unconstrained Maximization

In this section we consider a function $f: \mathbb{R}^N \to \mathbb{R}$.

**Definition M.J.1:** The vector $\bar{x} \in \mathbb{R}^N$ is a *local maximizer* of $f(\cdot)$ if there is an open neighborhood of $\bar{x}$, $A \subset \mathbb{R}^N$, such that $f(\bar{x}) \geq f(x)$ for every $x \in A$.[20] If $f(\bar{x}) \geq f(x)$ for every $x \in \mathbb{R}^N$ (i.e., if we can take $A = \mathbb{R}^N$), then we say that $\bar{x}$ is a *global maximizer* of $f(\cdot)$ (or simply a *maximizer*). The concepts of *local* and *global minimizers* are defined analogously.

In Figure M.J.1, we illustrate a local maximizer $\bar{x}$ and a local minimizer $\underline{x}$ (with open neighborhoods $A$ and $A'$, respectively) of a function for the case in which $N = 1$.

**Theorem M.J.1:** Suppose that $f(\cdot)$ is differentiable and that $\bar{x} \in \mathbb{R}^N$ is a local maximizer or local minimizer of $f(\cdot)$. Then

$$\frac{\partial f(\bar{x})}{\partial x_n} = 0 \qquad \text{for every } n, \tag{M.J.1}$$

or, in more concise notation,

$$\nabla f(\bar{x}) = 0. \tag{M.J.2}$$

**Proof:** Suppose that $\bar{x}$ is a local maximizer or local minimizer of $f(\cdot)$ but that $df(\bar{x})/\partial x_n = a > 0$ (the argument is analogous if $a < 0$). Denote by $e^n \in \mathbb{R}^N$ the vector having its $n$th entry equal to 1 and all other entries equal to 0 (i.e., having $e^n_n = 1$ and $e^n_h = 0$ for $h \neq n$). By the definition of a (partial) derivative, this means that there is an $\varepsilon > 0$ arbitrarily small such that $[f(\bar{x} + \varepsilon e^n) - f(\bar{x})]/\varepsilon > a/2 > 0$ and $[f(\bar{x} - \varepsilon e^n) - f(\bar{x})]/\varepsilon < -a/2$. Thus, $f(\bar{x} - \varepsilon e^n) < f(\bar{x}) < f(\bar{x} + \varepsilon e^n)$. In words: The function $f(\cdot)$ is locally increasing around $\bar{x}$ in the direction of the $n$th coordinate axis. But then $\bar{x}$ can be neither a local maximizer nor a local minimizer of $f(\cdot)$. Contradiction. ■

The conclusion of Theorem M.J.1 can be seen in Figure M.J.1: In the figure, we have $\partial f(\bar{x})/\partial x = 0$ and $\partial f(\underline{x})/\partial x = 0$.

A vector $\bar{x} \in \mathbb{R}^N$ such that $\nabla f(\bar{x}) = 0$ is called a *critical point*. By Theorem M.J.1, every local maximizer or local minimizer is a critical point. The converse, however,

20. An open neighborhood of $\bar{x}$ is an open set that includes $\bar{x}$.

does not hold. Consider for example, the function $f(x_1, x_2) = (x_1)^2 - (x_2)^2$ defined on $\mathbb{R}^2$. At the origin we have $\nabla f(0,0) = (0,0)$. Thus, the origin is a critical point, but it is neither a local maximizer nor a local minimizer of this function. To characterize local maximizers and local minimizers of $f(\cdot)$ more completely, we must look at second-order conditions.

**Theorem M.J.2:** Suppose that the function $f\colon \mathbb{R}^N \to \mathbb{R}$ is twice continuously differentiable and that $f(\bar{x}) = 0$.

(i) If $\bar{x} \in \mathbb{R}^N$ is a local maximizer, then the (symmetric) $N \times N$ matrix $D^2f(\bar{x})$ is negative semidefinite.
(ii) If $D^2f(\bar{x})$ is negative definite, then $\bar{x}$ is a local maximizer.

Replacing "negative" by "positive," the same is true for local minimizers.

**Proof:** The idea is as follows. For an arbitrary direction of displacement $z \in \mathbb{R}^N$ and scalar $\varepsilon$, a Taylor's expansion of the function $\phi(\varepsilon) = f(\bar{x} + \varepsilon z)$ around $\varepsilon = 0$ gives

$$\begin{aligned} f(\bar{x} + \varepsilon z) - f(\bar{x}) &= \varepsilon \nabla f(\bar{x}) \cdot z + \tfrac{1}{2}\varepsilon^2 z \cdot D^2 f(\bar{x}) z + \text{Remainder} \\ &= \tfrac{1}{2}\varepsilon^2 z \cdot D^2 f(\bar{x}) z + \text{Remainder}, \end{aligned}$$

where $\varepsilon \in \mathbb{R}_+$ and $(1/\varepsilon^2)$ Remainder is small if $\varepsilon$ is small. If $\bar{x}$ is a local maximizer, then for $\varepsilon$ small we must have $(1/\varepsilon^2)[f(\bar{x} + \varepsilon z) - f(\bar{x})] \le 0$, and so taking limits we get

$$z \cdot D^2 f(\bar{x}) z \le 0.$$

Similarly, if $z \cdot D^2 f(\bar{x}) z < 0$ for any $z \ne 0$, then $f(\bar{x} + \varepsilon z) < f(\bar{x})$ for $\varepsilon > 0$ small, and so $\bar{x}$ is a local maximizer. ■

In the borderline case in which $D^2 f(\bar{x})$ is negative semidefinite but not negative definite, we cannot assert that $\bar{x}$ is a local maximizer. Consider, for example, the function $f(x) = x^3$ whose domain is $\mathbb{R}$. Then $D^2 f(0)$ is negative semidefinite because $d^2 f(0)/dx = 0$, but $\bar{x} = 0$ is neither a local maximizer nor a local minimizer of this function.

Finally, when is a local maximizer $\bar{x}$ of $f(\cdot)$ (or, more generally, a critical point) automatically a global maximizer? Theorem M.J.3 tells us that a sufficient condition is the concavity of the objective function $f(\cdot)$.

**Theorem M.J.3:** Any critical point $\bar{x}$ of a concave function $f(\cdot)$ [i.e., any $\bar{x}$ satisfying $\nabla f(\bar{x}) = 0$] is a global maximizer of $f(\cdot)$.

**Proof:** Recall from Theorem M.C.1 that for a concave function we have $f(x) \le f(\bar{x}) + \nabla f(\bar{x}) \cdot (x - \bar{x})$ for every $x$ in the domain of the function. Since $\nabla f(\bar{x}) = 0$, this tells us that $\bar{x}$ is a global maximizer. ■

By analogous reasoning, any critical point of a *convex* function $f(\cdot)$ is a global *minimizer* of $f(\cdot)$.[21]

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21. In fact, this follows directly from Theorem M.J.3 because $\bar{x}$ is a global minimizer of $f(\cdot)$ if and only if it is a global maximizer of $-f(\cdot)$, and $-f(\cdot)$ is concave if and only if $f(\cdot)$ is convex.

# M.K Constrained Maximization

We start by considering the problem of maximizing a function $f(\cdot)$ under $M$ equality constraints. Namely, we study the problem

$$\begin{aligned} \operatorname*{Max}_{x \in \mathbb{R}^N} \quad & f(x) \\ \text{s.t. } & g_1(x) = \bar{b}_1 \\ & \vdots \\ & g_M(x) = \bar{b}_M, \end{aligned} \tag{M.K.1}$$

where the functions $f(\cdot), g_1(\cdot), \ldots, g_M(\cdot)$ are defined on $\mathbb{R}^N$ (or, more generally, on an open set $A \subset \mathbb{R}^N$). We assume that $N \geq M$; if $M \geq N$ there will generally be no points satisfying all of the constraints.

The set of all $x \in \mathbb{R}^N$ satisfying the constraints of problem (M.K.1) is denoted

$$C = \{x \in \mathbb{R}^N \colon g_m(x) = \bar{b}_m \quad \text{for} \quad m = 1, \ldots, M\}$$

and is called the *constraint set*. The definitions of a *local constrained* or a *global constrained maximizer* are parallel to those given in Definition M.J.1, except that we now consider only points $x$ that belong to the constraint set $C$. The feasible point $\bar{x} \in C$ is a *local constrained maximizer* in problem (M.K.1) if there exists an open neighborhood of $\bar{x}$, say $A \subset \mathbb{R}^N$, such that $f(\bar{x}) \geq f(x)$ for all $x \in A \cap C$, that is, if $\bar{x}$ solves problem (M.K.1) when we replace the condition $x \in \mathbb{R}^N$ by $x \in A$. The point $\bar{x}$ is a *global constrained maximizer* if it solves problem (M.K.1), that is, if $f(\bar{x}) \geq f(x)$ for all $x \in C$.

Our first result (Theorem M.K.1) states the first-order conditions for this constrained maximization problem.

**Theorem M.K.1:** Suppose that the objective and constraint functions of problem (M.K.1) are differentiable and that $\bar{x} \in C$ is a local constrained maximizer. Assume also that the $M \times N$ matrix

$$\begin{bmatrix} \dfrac{\partial g_1(\bar{x})}{\partial x_1} & \cdots & \dfrac{\partial g_1(\bar{x})}{\partial x_N} \\ & \ddots & \\ \dfrac{\partial g_M(\bar{x})}{\partial x_1} & \cdots & \dfrac{\partial g_M(\bar{x})}{\partial x_N} \end{bmatrix}$$

has rank $M$. (This is called the *constraint qualification*: It says that the constraints are independent at $\bar{x}$.) Then there are numbers $\lambda_m \in \mathbb{R}$, one for each constraint, such that

$$\frac{\partial f(\bar{x})}{\partial x_n} = \sum_{m=1}^{M} \lambda_m \frac{\partial g_m(\bar{x})}{\partial x_n} \quad \text{for every } n = 1, \ldots, N, \tag{M.K.2}$$

or, in more concise notation [letting $\lambda = (\lambda_1, \ldots, \lambda_M)$],

$$\nabla f(\bar{x}) = \sum_{m=1}^{M} \lambda_m \nabla g_m(\bar{x}). \tag{M.K.3}$$

The numbers $\lambda_m$ are referred to as *Lagrange multipliers*.

**Proof:** The role of the constraint qualification is to insure that $\bar{x}$ is also a local maximizer in the linearized problem

$$\begin{aligned} \underset{x \in \mathbb{R}^N}{\text{Max}} \quad & f(\bar{x}) + \nabla f(\bar{x}) \cdot (x - \bar{x}) \\ \text{s.t. } & \nabla g_1(\bar{x}) \cdot (x - \bar{x}) = 0 \\ & \vdots \\ & \nabla g_M(\bar{x}) \cdot (x - \bar{x}) = 0, \end{aligned}$$

in which the objective function and constraints have been linearized around the point $\bar{x}$. Thus, the constraint qualification guarantees the correctness of the following intuitively sensible statement: If $\bar{x}$ is a local constrained maximizer, then every direction of displacement $z \in \mathbb{R}^N$ having no first-order effect on the constraints, that is, satisfying $\nabla g_m(\bar{x}) \cdot z = 0$ for every $m$, must also have no first-order effect on the objective function, that is, must have $\nabla f(\bar{x}) \cdot z = 0$ (see also the discussion after the proof and Figure M.K.1). From now on we assume that this is true.

The rest is just a bit of linear algebra. Let $E$ be the $(M + 1) \times N$ matrix whose first row is $\nabla f(\bar{x})^{\mathrm{T}}$ and whose last $M$ rows are the vectors $\nabla g_1(\bar{x})^{\mathrm{T}}, \ldots, \nabla g_M(\bar{x})^{\mathrm{T}}$. By the implication of the constraint qualification cited above, we have $\{z \in \mathbb{R}^N\colon \nabla g_m(\bar{x}) \cdot z = 0$ for all $m\} = \{z \in \mathbb{R}^N\colon Ez = 0\}$. Hence, these two linear spaces have the same dimension $M$. Therefore, by a basic result of linear algebra, rank $E = M$. Hence, $\nabla f(\bar{x})$ must be a linear combination of the linearly independent set of gradients $\nabla g_1(\bar{x}), \ldots, \nabla g_M(\bar{x})$. This is exactly what (M.K.3) says. ■

In words, Theorem M.K.1 asserts that, at a local constrained maximizer $\bar{x}$, the gradient of the objective function is a linear combination of the gradients of the constraint functions. The indispensability of the constraint qualification is illustrated in Figure M.K.1. In the figure, we wish to maximize a linear function $f(x_1, x_2)$ in the constraint set $C = \{(x_1, x_2) \in \mathbb{R}^2\colon g_m(x_1, x_2) = \bar{b}_m$ for $m = 1, 2\}$ [the figure shows the loci of points satisfying $g_1(x_1, x_2) = \bar{b}_1$ and $g_2(x_1, x_2) = \bar{b}_2$, as well as the level sets of the function $f(\cdot)$]. While the point $\bar{x}$ is a global constrained maximum (it is the only vector in the constraint set!), we see that $\nabla f(\bar{x})$ is *not* spanned by the vectors $\nabla g_1(\bar{x})$ and $\nabla g_2(\bar{x})$ [i.e., it cannot be expressed as a linear combination of $\nabla g_1(\bar{x})$ and $\nabla g_2(\bar{x})$]. Note, however, that $\nabla g_1(\bar{x}) = -\nabla g_2(\bar{x})$, and so the constraint qualification

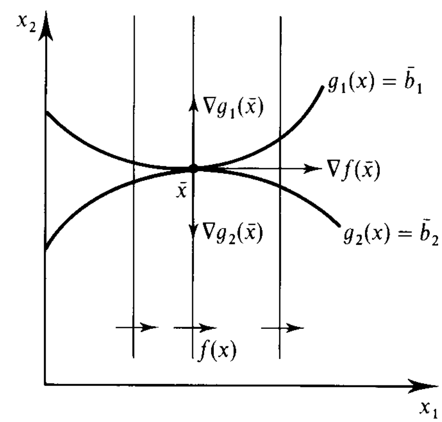



**Figure M.K.1**
Indispensability of the constraint qualification.

is violated. Observe that, indeed, $\bar{x}$ is *not* a local maximizer of $f(\bar{x}) + \nabla f(\bar{x})\cdot(x - \bar{x})$ [$= f(x)$ since $f(\cdot)$ is linear] in the linearized constraint set

$$C' = \{(x_1, x_2) \in \mathbb{R}^2 \colon \nabla g_m(\bar{x})\cdot(x - \bar{x}) = 0 \quad \text{for} \quad m = 1, 2\}.$$

Often the first-order conditions (M.K.2) or (M.K.3) are presented in a slightly different way. Given variables $x = (x_1, \ldots, x_N)$ and $\lambda = (\lambda_1, \ldots, \lambda_N)$, we can define the *Lagrangian function*

$$L(x, \lambda) = f(x) - \sum_m \lambda_m g_m(x).$$

Note that conditions (M.K.2) [or (M.K.3)] are the (unconstrained) first-order conditions of this function with respect to the variables $x = (x_1, \ldots, x_N)$. Similarly, the constraints $g(x) = 0$ are the first-order conditions of $L(\cdot, \cdot)$ with respect to the variables $\lambda = (\lambda_1, \ldots, \lambda_M)$. In summary, Theorem M.K.1 says that if $\bar{x}$ is a local constrained maximizer (and if the constraint qualification is satisfied), then for some values $\lambda_1, \ldots, \lambda_M$ all of the partial derivatives of the Lagrangian function are null; that is, $\partial L(\bar{x}, \lambda)/\partial x_n = 0$ for $n = 1, \ldots, N$ and $\partial L(\bar{x}, \lambda)/\partial \lambda_m = 0$ for $m = 1, \ldots, M$.

Theorem M.K.1 implies that if $\bar{x}$ is a local maximizer in problem (M.K.1), then the $N + M$ variables $(\bar{x}_1, \ldots, \bar{x}_N, \lambda_1, \ldots, \lambda_M)$ are a solution to the $N + M$ equations formed by (M.K.2) and $g_m(\bar{x}) = \bar{b}_m$ for $m = 1, \ldots, M$.

There is also a second-order theory associated with problem (M.K.1). Suppose that at $\bar{x}$ the constraint qualification is satisfied and that there are Lagrange multipliers satisfying (M.K.3). If $\bar{x}$ is a local maximizer, then

$$D_x^2 L(\bar{x}, \lambda) = D^2 f(\bar{x}) - \sum_m \lambda_m \nabla g_m(\bar{x})$$

is negative semidefinite on the subspace $\{z \in \mathbb{R}^N \colon \nabla g_m(\bar{x})\cdot z = 0 \text{ for all } m\}$. In the other direction, if the vector $\bar{x}$ is feasible (i.e., $\bar{x} \in C$) and satisfies the first-order conditions (M.K.2), and if $D_x^2 L(\bar{x}, \lambda)$ is negative definite on the subspace $\{z \in \mathbb{R}^N \colon \nabla g_m(\bar{x})\cdot z = 0$ for all $m\}$, then $\bar{x}$ is a local maximizer. These conditions can be verified using the determinantal tests provided in Theorem M.D.3.

Finally, note that a local constrained minimizer of $f(\cdot)$ is a local constrained maximizer of $-f(\cdot)$, and therefore Theorem M.K.1 and our discussion of second-order conditions above is also applicable to the characterization of local constrained minimizers.

### *Inequality Constraints*

We now generalize our analysis to problems that may have inequality constraints. The basic problem is therefore now

$$\begin{aligned} \underset{x \in \mathbb{R}^N}{\text{Max}} \quad & f(x) \\ \text{s.t. } & g_1(x) = \bar{b}_1 \\ & \vdots \\ & g_M(x) = \bar{b}_M \\ & h_1(x) \le \bar{c}_1 \\ & \vdots \\ & h_K(x) \le \bar{c}_K, \end{aligned} \tag{M.K.4}$$

where every function is defined on $\mathbb{R}^N$ (or an open set $A \subset \mathbb{R}^N$). We assume that $N \geq M + K$. It is of course possible to have $M = 0$ (no equality constraints) or $K = 0$ (no inequality constraints).

We again denote the constraint set by $C \subset \mathbb{R}^N$, and the meaning of a local constrained maximizer or a global constrained maximizer is unaltered from above.

We now say that the constraint qualification is satisfied at $\bar{x} \in C$ if the constraints *that hold at* $\bar{x}$ *with equality* are independent; that is, if the vectors in $\{\nabla g_m(\bar{x}): m = 1, \ldots, M\} \cup \{\nabla h_k(\bar{x}): h_k(\bar{x}) = \bar{c}_k\}$ are linearly independent.

Theorem M.K.2 presents the first-order conditions for this problem. All of the functions involved are assumed to be differentiable.

**Theorem M.K.2:** (*Kuhn–Tucker Conditions*) Suppose that $\bar{x} \in C$ is a local maximizer of problem (M.K.4). Assume also that the constraint qualification is satisfied. Then there are multipliers $\lambda_m \in \mathbb{R}$, one for each equality constraint, and $\lambda_k \in \mathbb{R}_+$, one for each inequality constraint, such that:[22]

(i) For every $n = 1, \ldots, N$,

$$\frac{\partial f(\bar{x})}{\partial x_n} = \sum_{m=1}^{M} \lambda_m \frac{\partial g_m(\bar{x})}{\partial x_n} + \sum_{k=1}^{K} \lambda_k \frac{\partial h_k(\bar{x})}{\partial x_n}, \tag{M.K.5}$$

or, in more concise notation,

$$\nabla f(\bar{x}) = \sum_{m=1}^{M} \lambda_m \nabla g_m(\bar{x}) + \sum_{k=1}^{K} \lambda_k \nabla h_k(\bar{x}). \tag{M.K.6}$$

(ii) For every $k = 1, \ldots, K$,

$$\lambda_k(h_k(\bar{x}) - \bar{c}_k) = 0, \tag{M.K.7}$$

i.e., $\lambda_k = 0$ for any constraint $k$ that does not hold with equality.

**Proof:** We illustrate the proof of the result for the case in which there are only inequality constraints (i.e., $M = 0$).

As with the case of equality constraints, the role of the constraint qualification is to insure that $\bar{x}$ remains a local maximizer in the problem linearized around $\bar{x}$. More specifically, we assume from now on that the following is true: Any direction of displacement $z \in \mathbb{R}^N$ that satisfies the constraints to first order [i.e., such that $\nabla h_k(\bar{x}) \cdot z \leq 0$ for every $k$ with $h_k(\bar{x}) = \bar{c}_k$] must not create a first-order increase in the objective function, that is, must have $\nabla f(\bar{x}) \cdot z \leq 0$.

In Figure M.K.2 we represent a problem with two variables and two constraints for which the logic of the result is illustrated and made plausible. The Kuhn–Tucker theorem says that if $\bar{x}$ is a local maximizer then $\nabla f(\bar{x})$ must be in the cone

$$\Gamma = \{y \in \mathbb{R}^2: y = \lambda_1 \nabla h_1(\bar{x}) + \lambda_2 \nabla h_2(\bar{x}) \quad \text{for some} \quad (\lambda_1, \lambda_2) \geq 0\}$$

depicted in the figure; that is, $\nabla f(\bar{x})$ must be a nonnegative linear combination of $\nabla h_1(\bar{x})$ and $\nabla h_2(\bar{x})$. Suppose now that $\bar{x}$ is a local maximizer. If starting from $\bar{x}$ we move along the boundary of the constraint set to any point $(\bar{x}_1 + z_1, \bar{x}_2 + z_2)$ with $z_1 < 0$ and $z_2 > 0$, then in the situation depicted we would have

$$h_1(\bar{x}_1 + z_1, \bar{x}_2 + z_2) = \bar{c}_1, \qquad h_2(\bar{x}_1 + z_1, \bar{x}_2 + z_2) < \bar{c}_2,$$

22. By convention, if there are no constraints (i.e., if $M = K = 0$), then the right-hand side of (M.K.5) is zero.

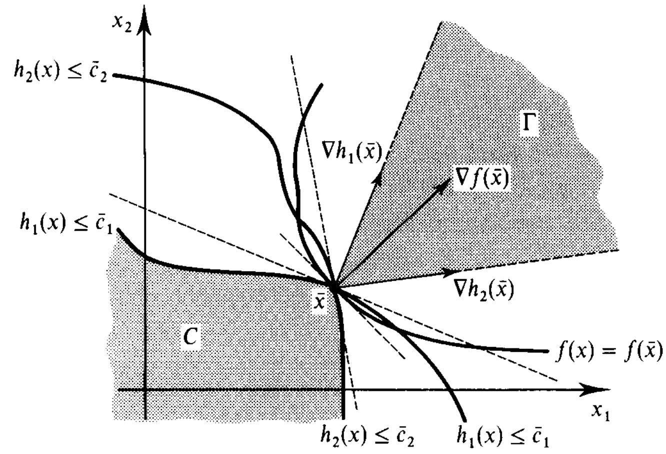



**Figure M.K.2**
Necessity of the Kuhn–Tucker conditions.

and $f(\bar{x}_1 + z_1, \bar{x}_2 + z_2) \leq f(\bar{x})$. Taking limits we conclude that in the direction $z$ such that $\nabla h_1(\bar{x}) \cdot z = 0$ and $\nabla h_2(\bar{x}) \cdot z < 0$, we have $\nabla f(\bar{x}) \cdot z \leq 0$. Geometrically this means that the vector $\nabla f(\bar{x})$ must lie below the vector $\nabla h_1(\bar{x})$, as is shown in the figure. By similar reasoning (moving along the boundary of the constraint set $C$ in the opposite direction), if $\bar{x}$ is a local constrained maximizer the vector $\nabla f(\bar{x})$ must lie above the vector $\nabla h_2(\bar{x})$. Hence, $\nabla f(\bar{x})$ must lie in the cone $\Gamma$. This is precisely what the Kuhn–Tucker conditions require in this case.

The above intuition can be extended to the general case. Suppose that all the constraints are binding at $\bar{x}$ (if a constraint $k$ is not binding, put $\lambda_k = 0$ and drop it from the list). We must show that $\nabla f(\bar{x})$ belongs to the convex cone

$$\Gamma = \left\{ y \in \mathbb{R}^N \colon y = \sum_k \lambda_k \nabla h_k(\bar{x}) \quad \text{for some} \quad (\lambda_1, \ldots, \lambda_K) \geq 0 \right\} \subset \mathbb{R}^N.$$

Assume for a moment that this is not so, that is, that $\nabla f(\bar{x}) \notin \Gamma$. Then, by the separating hyperplane theorem (Theorem M.G.2), there exists a nonzero vector $z \in \mathbb{R}^N$ and a number $\beta \in \mathbb{R}$ such that $\nabla f(\bar{x}) \cdot z > \beta$ and $y \cdot z < \beta$ for every $y \in \Gamma$. Since $0 \in \Gamma$ we must have $\beta > 0$. Hence, $\nabla f(\bar{x}) \cdot z > 0$. Also, for any $y \in \Gamma$ we have $\theta y \in \Gamma$ for all $\theta \geq 0$. But then $\theta y \cdot z < \beta$ can hold for all $\theta$ (arbitrarily large) only if $y \cdot z \leq 0$. We conclude therefore that $\nabla f(\bar{x}) \cdot z > 0$ and $\nabla h_k(\bar{x}) \cdot z \leq 0$ for all $k$, which contradicts the linearization implication of the constraint qualification. ■

It is common in applications that a constraint takes the form of a nonnegativity requirement on some variable $x_n$; that is, $x_n \geq 0$. In this case, the appropriate first-order conditions require only a small modification of those above. In particular, we need only change the first-order condition for $x_n$ to

$$\frac{\partial f(\bar{x})}{\partial x_n} \leq \sum_{m=1}^{M} \lambda_m \frac{\partial g_m(\bar{x})}{\partial x_n} + \sum_{k=1}^{K} \lambda_k \frac{\partial h_m(\bar{x})}{\partial x_n}, \qquad \text{with equality if } \bar{x}_n > 0. \quad \text{(M.K.8)}$$

To see why this is so, suppose that we explicitly introduced this nonnegativity requirement as our $(K + 1)$th inequality constraint [i.e., $h_{K+1}(x) = -x_n \leq 0$] and let $\lambda_{K+1} \geq 0$ be the corresponding multiplier. Note that $\lambda_{K+1}(\partial h_{K+1}(\bar{x})/\partial x_n) = -\lambda_{K+1}$ and $\partial h_{K+1}(\bar{x})/\partial x_{n'} = 0$ for $n' \neq n$. Thus, if we apply condition (M.K.5) of Theorem M.K.2, the only modification to the first-order conditions is that the first-order condition for $x_n$ is now

$$\frac{\partial f(\bar{x})}{\partial x_n} = \sum_{m=1}^{M} \lambda_m \frac{\partial g_m(\bar{x})}{\partial x_n} + \sum_{k=1}^{K} \lambda_k \frac{\partial h_m(\bar{x})}{\partial x_n} - \lambda_{K+1},$$

and we have the added condition that

$$-\lambda_{K+1} x_n = 0.$$

But these two conditions are exactly equivalent to condition (M.K.8). Given the simplicity of the adjustment required to take account of nonnegativity constraints, it is customary in applications not to explicitly introduce the nonnegativity constraint and its associated multiplier, but rather simply to modify the usual first-order conditions as in (M.K.8).

Note also that any constraint of the form $h_k(x) \geq \bar{c}_k$ can be written as $-h_k(x) \leq -\bar{c}_k$. Using this fact, we see that Theorem M.K.2 extends to constraints of the form $h_k(x) \geq \bar{c}_k$. The only modification is that the sign restriction on the multiplier of constraint $k$ is now $\lambda_k \leq 0$. Similarly, because minimizing the function $f(\cdot)$ is equivalent to maximizing the function $-f(\cdot)$, Theorem M.K.2 applies also to local minimizers, with the only change being that the sign restriction on all of the multipliers is now $(\lambda_1, \ldots, \lambda_M) \leq 0$ [assuming that the constraints are all still written as in problem (M.K.4)].

The second-order conditions for the inequality problem (M.K.4) are exactly the same as those already mentioned for the equality problem (M.K.1). The only adjustment is that the constraints that count are those that bind, that is, those that hold with equality at the point $\bar{x}$ under consideration.

Suppose that a vector $x \in C$ satisfies the Kuhn–Tucker conditions, that is, conditions (i) and (ii) in Theorem M.K.2. When can we say that $x$ is a global maximizer? Theorem M.K.3 gives a useful set of conditions.

**Theorem M.K.3:** Suppose that there are no equality constraints (i.e., $M = 0$) and that every inequality constraint $k$ is given by a quasiconvex function $h_k(\cdot)$.[23] Suppose also that the objective function satisfies

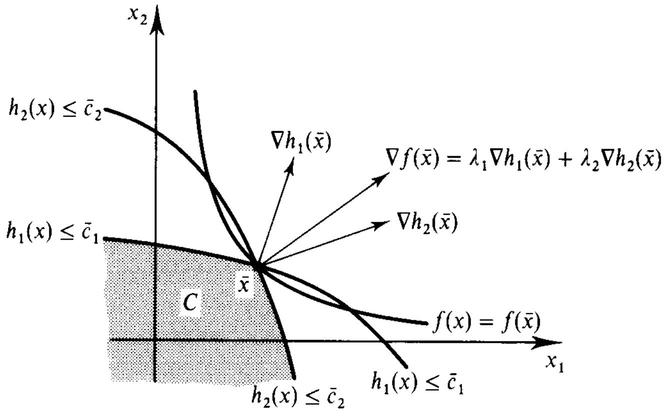
$$\nabla f(x) \cdot (x' - x) > 0 \qquad \text{for any } x \text{ and } x' \text{ with } f(x') > f(x). \qquad \text{(M.K.9)}$$

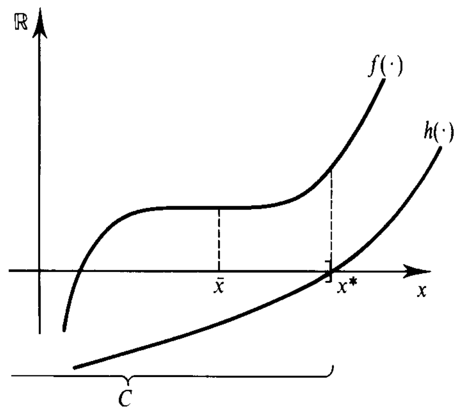
Then if $\bar{x} \in C$ satisfies the Kuhn–Tucker conditions [conditions (i) and (ii) of Theorem M.K.2], and if the constraint qualification holds at $\bar{x}$, it follows that $\bar{x}$ is a global maximizer.[24]

**Proof:** Suppose that this is not so, that is, that $f(x) > f(\bar{x})$ for some $x \in \mathbb{R}^N$ satisfying $h_k(x) \leq \bar{c}_k$ for every $k$. Denote $z = x - \bar{x}$. Then, by (M.K.9) we have $\nabla f(\bar{x}) \cdot z > 0$. If $\lambda_k > 0$, then the Kuhn–Tucker conditions imply that $h_k(\bar{x}) = \bar{c}_k$. Moreover, since $h_k(\cdot)$ is quasiconvex and $h_k(x) \leq \bar{c}_k = h_k(\bar{x})$, it follows that $\nabla h_k(\bar{x}) \cdot z \leq 0$. Hence, we have both $\nabla f(\bar{x}) \cdot z > 0$ and $\sum_k \lambda_k \nabla h_k(\bar{x}) \cdot z \leq 0$, which contradicts the Kuhn–Tucker conditions because these require that $\nabla f(\bar{x}) = \sum_k \lambda_k \nabla h_k(\bar{x})$. ■

23. More generally, equality constraints are permissible if they are linear.

24. If instead we have $\nabla f(x) \cdot (x' - x) < 0$ whenever $f(x') < f(x)$ and the multipliers have the nonpositive sign that corresponds to a minimization problem, then $\bar{x}$ is a global minimizer.




Note that condition (M.K.9) of Theorem M.K.3 is satisfied if $f(\cdot)$ is concave or if $f(\cdot)$ is quasiconcave and $\nabla f(x) \neq 0$ for all $x \in \mathbb{R}^N$. The condition that the constraint functions $h_1(\cdot), \ldots, h_K(\cdot)$ are quasiconvex implies that the constraint set $C$ is convex (check this).[25] In Figure M.K.3, we illustrate the theorem for a case in which $N = K = 2$, $M = 0$, and $f(\cdot)$ is a quasiconcave function with $\nabla f(x) \neq 0$ for all $x$.

The indispensability of condition (M.K.9) in Theorem M.K.3 is shown in Figure M.K.4. There we have $N = M = 1$, and the quasiconcave function $f(\cdot)$ is being maximized on the constraint set $C = \{x \in \mathbb{R}: h(x) \le 0\}$. In the figure, the point $\bar{x}$ satisfies the Kuhn–Tucker conditions for a multiplier value of $\lambda = 0$, but $\bar{x}$ is not a global maximizer of $f(\cdot)$ on $C$ (the point $x^*$ is the global constrained maximizer). Note, however, that condition (M.K.10) is violated when $x = \bar{x}$ and $x' = x^*$.

We observe finally in Theorem M.K.4 an important implication of the constraint set $C$ being convex and the objective function $f(\cdot)$ being strictly quasiconcave.

**Figure M.K.3 (left)** With quasiconvex constraint functions and a quasiconcave objective function satisfying $\nabla f(x) \neq 0$ for all $x$, satisfaction of the Kuhn–Tucker conditions at $\bar{x}$ implies that $\bar{x}$ is a global constrained maximizer.

**Figure M.K.4 (right)** The necessity of condition (M.K.10) for Theorem M.K.3.

**Theorem M.K.4:** Suppose that in problem (M.K.4) the constraint set $C$ is convex and the objective function $f(\cdot)$ is strictly quasiconcave. Then there is a unique global constrained maximizer.[26]

**Proof:** If $x$ and $x' \neq x$ were both global constrained maximizers, then the point $x'' = \alpha x + (1 - \alpha)x'$ for $\alpha \in (0, 1)$ would be feasible (i.e., $x'' \in C$) and by the strict quasiconcavity of $f(\cdot)$, would yield a higher value of $f(\cdot)$ [i.e., $f(x'') > f(x) = f(x')$]. ∎

---

Suppose that in the case in which only inequality constraints are present we denote by $C_{-k}$ the relaxed constraint set arising when the $k$th inequality constraint is dropped. The following two facts are often useful in applications.

(i) If $f(\bar{x}) \ge f(x)$ for all $x \in C_{-k}$ and if $h_k(\bar{x}) \le \bar{c}_k$, then $\bar{x}$ is a global constrained maximizer in problem (M.K.4). That is, if we solve a constrained optimization problem ignoring a constraint, and the solution we obtain satisfies this omitted constraint as well, then it must be a solution to the fully constrained problem. This follows simply from the fact that $C \subset C_{-k}$, and so optimizing $f(\cdot)$ on $C$ can at best yield the same value of $f(\cdot)$ as optimizing it on $C_{-k}$.

---

25. Also, under the conditions of the theorem, a sufficient constraint qualification is that the constraint set $C$ should have a nonempty interior.

26. When $f(\cdot)$ is quasiconcave, but not strictly so, a similar argument allows us to conclude that the set of maximizers is convex.

(ii) Suppose that all of the constraint functions $h_1(\cdot), \ldots, h_K(\cdot)$ are continuous and quasiconvex and that condition (M.K.9) holds. Then if $\bar{x}$ is a solution to problem (M.K.4) in which the $k$th constraint is not binding [i.e., if $h_k(\bar{x}) < \bar{c}_k$], we have $f(\bar{x}) \geq f(x)$ for all $x \in C_{-k}$. That is, under the stated assumptions, if a constraint is not binding at a solution to problem (M.K.4), then ignoring it altogether should have no effect on the solution. To see this, suppose otherwise; i.e., that there is an $x' \in C_{-k}$ such that $f(x') > f(\bar{x})$. Then because the constraint functions $h_1(\cdot), \ldots, h_K(\cdot)$ are quasiconvex, we know that the point $x(\alpha) = \alpha x' + (1 - \alpha)\bar{x}$ is an element of $C_{-k}$ for all $\alpha \in [0, 1]$. Moreover, since the $k$th constraint is not binding at $\bar{x}$, there is an $\bar{\alpha} > 0$ such that $h_k(x(\alpha)) < \bar{c}_k$ for all $\alpha < \bar{\alpha}$. Hence, $x(\alpha) \in C$ for all $\alpha < \bar{\alpha}$. But the derivative of $f(x(\alpha))$ at $\alpha = 0$ is $\nabla f(x) \cdot (x' - \bar{x}) > 0$ [recall that condition (M.K.9) holds and that, by assumption $f(x') > f(\bar{x})$]. Therefore, there must be a point $x(\alpha) \in C$ such that $f(x(\alpha)) > f(\bar{x})$—a contradiction to $\bar{x}$ being a global constrained maximizer in problem (M.K.4).

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## *Comparative Statics*

In our previous discussion we have treated the parameters $\bar{b} = (\bar{b}_1, \ldots, \bar{b}_M)$ and $\bar{c} = (\bar{c}_1, \ldots, \bar{c}_K)$ of problem M.K.4 as given. We will now let them vary.

Suppose that $(b, c) \in \mathbb{R}^{M+K}$ are parameters for which problem (M.K.4) has some solution $\bar{x}(b, c)$ and denote the value of this solution by $v(b, c) = f(\bar{x}(b, c))$. Under fairly general conditions (see the small type at the end of this section), the value $v(b, c)$ depends continuously on the parameters $(b, c)$.

Theorem M.K.5 provides an interpretation for the Lagrange multipliers as the "shadow prices" of the constraints.

**Theorem M.K.5:** Suppose that in an open neighborhood of $(\bar{b}, \bar{c})$ the set of binding constraints remains unaltered and that $v(b, c)$ is differentiable at $(\bar{b}, \bar{c})$.[27] Then for every $m = 1, \ldots, M$ and $k = 1, \ldots, K$ we have

$$\frac{\partial v(\bar{b}, \bar{c})}{\partial b_m} = \lambda_m \quad \text{and} \quad \frac{\partial v(\bar{b}, \bar{c})}{\partial c_k} = \lambda_k.$$

**Proof:** This is a particular case of the envelope theorem (Theorem M.L.1) to be presented in the next section. ■

---

Consider a more general optimization problem. We maximize a function $f\colon \mathbb{R}^N \to \mathbb{R}$ subject to $x \in C(q)$ where $C(q)$ is a nonempty constraint set and $q = (q, \ldots, q_S)$ belongs to an admissible set of parameters $Q \subset \mathbb{R}^S$. Suppose that $f(\cdot)$ is continuous and that $C(q)$ is compact for every $q \in Q$. Then we know [from Theorem M.F.2, part (ii)] that the maximum problem has at least one solution. Denote by $x(q) \subset C(q)$ the *set* of solutions corresponding to $q$ and by $v(q)$ [$= f(x)$ for any $x \in x(q)$] the associated maximum value. Theorem M.K.6 concerns the continuity of $x(\cdot)$ and $v(\cdot)$.

**Theorem M.K.6:** (*Theorem of the Maximum*) Suppose that the constraint correspondence $C\colon Q \to \mathbb{R}^N$ is continuous (see Section M.H) and that $f(\cdot)$ is a continuous function. Then the maximizer correspondence $x\colon Q \to \mathbb{R}^N$ is upper hemicontinuous and the value function $v\colon Q \to \mathbb{R}$ is continuous.

---

27. These are simplifying assumptions; a similar result holds more generally but requires the use of directional derivatives at points of nondifferentiability of the function $v(\cdot, \cdot)$.

The result cannot be improved upon. Suppose that we maximize $x_1 + x_2$ subject to $x_1 \in [0, 1]$, $x_2 \in [0, 1]$, and $q_1 x_1 + q_2 x_2 \leq q_1 q_2$ for $q = (q_1, q_2) \in Q = (0, 1)^2$. Then the maximizer correspondence is given by

$$x(q) = \{(q_2, 0)\} \qquad \text{if } q_1 < q_2,$$

$$x(q) = \{(0, q_1)\} \qquad \text{if } q_2 < q_1,$$

and

$$x(q) = \{(x_1, x_2) \in [0, 1]^2 : x_1 + x_2 = q_1\} \qquad \text{if } q_1 = q_2.$$

Both the objective function and the constraint correspondence are continuous (you should check the latter). In accordance with Theorem M.K.6, $x(\cdot)$ is upper hemicontinuous. But it is not continuous (there is an explosion along the line $q_1 = q_2$). On the other hand, suppose that we take $Q = [0, 1]^2$. Then the conclusion of the theorem fails: the maximizer correspondence is not upper hemicontinuous [we have $x(2\varepsilon, \varepsilon) = \{(0, 2\varepsilon)\}$, but $x(0, 0) = \{(1, 1)\}$]. However, the assumptions also fail: at $q = (0, 0)$ the vector $(1, 1)$ belongs to the constraint set, but at $q = (\varepsilon, \varepsilon)$ no vector $x$ with $x_1 + x_2 > \varepsilon$ belongs to the constraint set. Hence the constraint correspondence is not continuous once extended to $Q = [0, 1]^2$.

# M.L The Envelope Theorem

In this section, we return to the problem of maximizing a function $f(\cdot)$ under constraints, but we suppose that we want to keep track of some parameters $q = (q_1, \ldots, q_S) \in \mathbb{R}^S$ entering the objective function or the constraints of the problem. In particular, we now write the maximization problem as

$$\begin{aligned} \underset{x \in \mathbb{R}^N}{\text{Max}} \quad & f(x; q) \\ \text{s.t.} \quad & g_1(x; q) = \bar{b}_1 \\ & \vdots \\ & g_M(x; q) = \bar{b}_M. \end{aligned} \tag{M.L.1}$$

We denote by $v(\cdot)$ the *value function* of problem (M.L.1); that is, $v(q)$ is the value attained by $f(\cdot)$ at a solution to problem (M.L.1) when the parameter vector is $q$. To be specific, we suppose that $v(q)$ is well-defined in the neighborhood of some reference parameter vector $\bar{q} \in \mathbb{R}^S$. It is then natural to investigate the marginal effects of changes in $q$ on the value $v(q)$. The *envelope theorem* addresses this matter.[28]

It will be convenient from now on to assume that, at least locally (i.e., for values of $q$ close to $\bar{q}$), the solution to problem (M.L.1) is a (differentiable) function $x(q)$. We can then write $v(q) = f(x(q); q)$.

To start with the simplest case, suppose that there is a single variable and a single parameter (i.e., $N = K = 1$) and that there are no constraints (i.e., $M = 0$). Then, by the chain rule,

$$\frac{dv(\bar{q})}{dq} = \frac{\partial f(x(\bar{q}); \bar{q})}{\partial q} + \frac{\partial f(x(\bar{q}); \bar{q})}{\partial x} \frac{dx(\bar{q})}{dq}. \tag{M.L.2}$$

28. Formally, we are presenting a case with equality constraints. But note that as long as in a neighborhood of the parameter vector under consideration the set of binding constraints does not change, the discussion applies automatically to the case with inequality constraints.

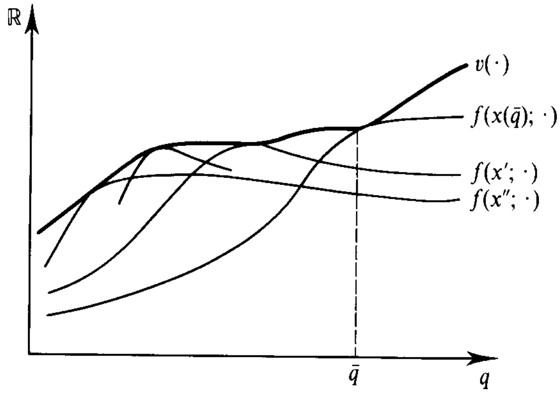



**Figure M.L.1**
The envelope theorem.

But note, and this is a key observation, that by the first-order conditions for unconstrained maximization (see Section M.J), we must have $\partial f(x(\bar{q}); \bar{q})/\partial x = 0$. Therefore, (M.L.2) simplifies to

$$\frac{dv(\bar{q})}{dq} = \frac{\partial f(x(\bar{q}), \bar{q})}{\partial q}. \tag{M.L.3}$$

That is, the fact that $x(q)$ is determined by maximizing the function $f(\cdot; q)$ has the implication that in computing the first-order effects of changes in $q$ on the maximum value, we can equally well assume that the maximizer will not adjust: The only effect of any consequence is the direct effect.

This result is illustrated in Figure M.L.1, which also motivates the use of the term "envelope." In the figure we represent the function $f(x; \cdot)$ for different values of $x$. Because at every $q$ we have $v(q) = \text{Max}_x\, f(x; q)$, the value function $v(\cdot)$ is given by the upper envelope of these functions. Suppose now that we consider a fixed $\bar{q}$. Then, denoting $\bar{x} = x(\bar{q})$, we have $f(\bar{x}; q) \le v(q)$ for all $q$, and $f(\bar{x}; \bar{q}) = v(\bar{q})$. Hence, the graph of $f(\bar{x}; \cdot)$ lies weakly below the graph of $v(\cdot)$ and touches it when $q = \bar{q}$. So the two graphs have the same slope at that point. This is precisely what condition (M.L.3) says.

We now state the general envelope theorem for a problem with any number of variables, parameters, and constraints. As we will see, its conclusion is similar to (M.L.3), except that Lagrange multipliers play an important role.

**Theorem M.L.1:** (*Envelope Theorem*) Consider the value function $v(q)$ for the problem (M.L.1). Assume that it is differentiable at $\bar{q} \in \mathbb{R}^S$ and that $(\lambda_1, \ldots, \lambda_M)$ are values of the Lagrange multipliers associated with the maximizer solution $x(\bar{q})$ at $\bar{q}$. Then[29]

$$\frac{\partial v(\bar{q})}{\partial q_s} = \frac{\partial f(x(\bar{q}); \bar{q})}{\partial q_s} - \sum_{m=1}^{M} \lambda_m \frac{\partial g_m(x(\bar{q}); \bar{q})}{\partial q_s} \quad \text{for } s = 1, \ldots, S, \tag{M.L.4}$$

29. If we have a case with inequality constraints in which the set of binding constraints remains unaltered in a neighborhood of $\bar{q}$, then expressions (M.L.4) and (M.L.5) are still valid: Accounting for the nonbinding constraints will have no effect either on the left-hand side or on the right-hand side (because its associated multipliers are zero).

or, in matrix notation,

$$\nabla v(\bar{q}) = \nabla_q f(x(\bar{q}); \bar{q}) - \sum_{m=1}^{M} \lambda_m \nabla_q g_m(x(\bar{q}); \bar{q}). \tag{M.L.5}$$

**Proof:** We proceed as in the case of a single variable and no constraints. Let $x(\cdot)$ be the maximizer function. Then $v(q) = f(x(q); q)$ for all $q$, and therefore, using the chain rule, we have

$$\frac{\partial v(\bar{q})}{\partial q_s} = \frac{\partial f(x(\bar{q}); \bar{q})}{\partial q_s} + \sum_{n=1}^{N} \left( \frac{\partial f(x(\bar{q}); \bar{q})}{\partial x_n} \frac{\partial x_n(\bar{q})}{\partial q_s} \right).$$

The first-order conditions (M.K.2) tell us that

$$\frac{\partial f(x(\bar{q}); \bar{q})}{\partial x_n} + \sum_{m=1}^{M} \lambda_m \frac{\partial g_m(x(\bar{q}); \bar{q})}{\partial x_n}.$$

Hence (switching the order of summation as we go),

$$\frac{\partial v(\bar{q})}{\partial q_s} = \frac{\partial f(x(\bar{q}); \bar{q})}{\partial q_s} + \sum_{m=1}^{M} \lambda_m \sum_{n=1}^{N} \left( \frac{\partial g_m(x(\bar{q}); \bar{q})}{\partial x_n} \frac{\partial x_n(\bar{q})}{\partial q_s} \right).$$

Moreover, since $g_m(x(q); q) = \bar{b}_m$ for all $q$, we have

$$\sum_{n=1}^{N} \left( \frac{\partial g_m(x(\bar{q}); \bar{q})}{\partial x_n} \frac{\partial x_n(\bar{q})}{\partial q_s} \right) = -\frac{\partial g_m(\bar{x}; \bar{q})}{\partial q_s} \quad \text{for all } m = 1, \ldots, M.$$

Combining, we get (M.L.4). ■

# M.M Linear Programming

Linear programming problems constitute the special cases of constrained maximization problems for which both the constraints and the objective function are linear in the variables $(x_1, \ldots, x_N)$.

A general linear programming problem is typically written in the form

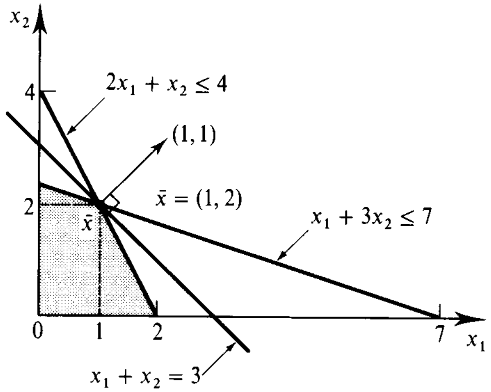
$$\begin{aligned} \underset{(x_1,\ldots,x_N) \geq 0}{\text{Max}} \quad & f_1 x_1 + \cdots + f_N x_N \\ \text{s.t. } & a_{11}x_1 + \cdots + a_{1N}x_N \leq c_1 \\ & \vdots \\ & a_{K1}x_1 + \cdots + a_{KN}x_N \leq c_K, \end{aligned} \tag{M.M.1}$$

or, in matrix notation,

$$\begin{aligned} \underset{x \in \mathbb{R}^N_+}{\text{Max}} \quad & f \cdot x \\ \text{s.t. } & Ax \leq c, \end{aligned}$$

where $A$ is the $K \times N$ matrix with generic entry $a_{kn}$, and $f \in \mathbb{R}^N$, $x \in \mathbb{R}^N$, and $c \in \mathbb{R}^K$ are (column) vectors.[30]

30. We say that this is the general form of the linear programming problem because, first, an equality constraint $a \cdot x = b$ can always be expressed as two inequality constraints ($a \cdot x \leq b$ and $-a \cdot x \geq b$) and, second, a variable $x_n$ that is unrestricted in sign can always be replaced by the difference of two variables $(x_{n+} - x_{n-})$, each restricted to be nonnegative.



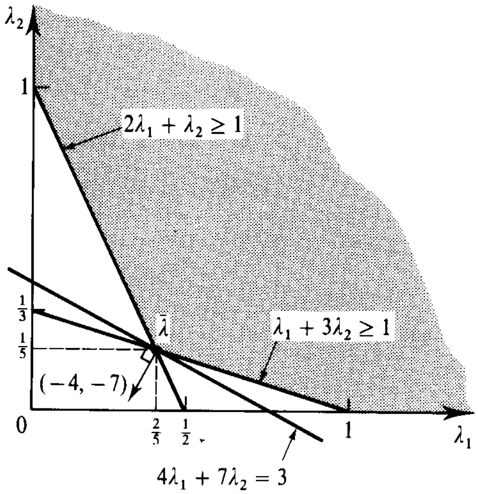




**Figure M.M.1 (left)**
A linear programming problem (the primal).

**Figure M.M.2 (right)**
A linear programming problem (the dual).

Figure M.M.1 represents a linear programming problem with $N = 2$, the two constraints $2x_1 + x_2 \le 4$ and $x_1 + 3x_2 \le 7$, and the objective function $x_1 + x_2$.

A most interesting fact about the linear programming problem (M.M.1) is that with it we can associate another linear programming problem, called the *dual* problem, that has the form of a minimization problem with $K$ variables (one for each constraint of the original, or *primal*, problem) and $N$ constraints (one for each variable of the primal problem):

$$\begin{aligned} \operatorname*{Min}_{(\lambda_1,\dots,\lambda_K)\ge 0} \quad & c_1\lambda_1 + \cdots + c_K\lambda_K \\ \text{s.t. } & a_{11}\lambda_1 + \cdots + a_{K1}\lambda_K \ge f_1 \\ & \vdots \\ & a_{1N}\lambda_1 + \cdots + a_{KN}\lambda_K \ge f_N, \end{aligned} \tag{M.M.2}$$

or, in matrix notation,

$$\begin{aligned} \operatorname*{Max}_{\lambda \in \mathbb{R}^K_+} \quad & c\cdot\lambda \\ \text{s.t. } & A^{\mathrm{T}}\lambda \ge f, \end{aligned}$$

where $\lambda \in \mathbb{R}^K$ is a column vector.

Figure M.M.2 represents the dual problem associated with Figure M.M.1. The constraints are now $2\lambda_1 + \lambda_2 \ge 1$ and $\lambda_1 + 3\lambda_2 \ge 1$, and the objective function is now $4\lambda_1 + 7\lambda_2$.

Suppose that $x \in \mathbb{R}^N_+$ and $\lambda \in \mathbb{R}^K_+$ satisfy, respectively, the constraints of the primal and the dual problems. Then

$$f\cdot x \le (A^{\mathrm{T}}\lambda)\cdot x = \lambda\cdot(Ax) \le \lambda\cdot c = c\cdot\lambda. \tag{M.M.3}$$

Thus, the solution value to the primal problem can be no larger than the solution value to the dual problem. The *duality theorem of linear programming*, now to be stated, says that these values are actually equal. The key for an understanding of this fact is that, as the notation suggests, the dual variables $(\lambda_1, \dots, \lambda_K)$ have the interpretation of Lagrange multipliers.

**Theorem M.M.1:** (*Duality Theorem of Linear Programming*) Suppose that the primal problem (M.M.1) attains a maximum value $v \in \mathbb{R}$. Then $v$ is also the minimum value attained by the dual problem (M.M.2).

**Proof:** Let $\bar{x} \in \mathbb{R}^N$ be a maximizer vector for problem (M.M.1). Denote by $\bar{\lambda} = (\bar{\lambda}_1, \ldots, \bar{\lambda}_K) \geq 0$ the Lagrange multipliers associated with this problem (see Theorem M.K.2).[31] Formally, we regard $\bar{\lambda}$ as a column vector. Then, applying Theorem M.K.2, we have

$$A^{\mathrm{T}}\bar{\lambda} = f \qquad \text{and} \qquad \bar{\lambda}\cdot(c - A\bar{x}) = 0.$$

Hence, $\bar{\lambda}$ satisfies the constraints of the dual problem (since $A^{\mathrm{T}}\bar{\lambda} \geq f$) and

$$c\cdot\bar{\lambda} = \bar{\lambda}\cdot c = \bar{\lambda}\cdot A\bar{x} = (A^{\mathrm{T}}\bar{\lambda})\cdot\bar{x} = f\cdot\bar{x}. \tag{M.M.4}$$

Now, by (M.M.3), we know that $c\cdot\lambda \geq f\cdot\bar{x}$ for all $\lambda \in \mathbb{R}^K_+$ such that $A^{\mathrm{T}}\lambda \geq f$. Therefore $c\cdot\bar{\lambda} \leq c\cdot\lambda$ if $A^{\mathrm{T}}\lambda \geq f$. So (M.M.4) tell us that, in fact, $\bar{\lambda}$ solves the dual problem (M.M.2) and therefore the value of the dual problem, $c\cdot\bar{\lambda}$, equals $f\cdot\bar{x}$, the value of the primal problem. ∎

We can verify the duality theorem for the primal and dual problems of Figures M.M.1 and M.M.2. The maximizer vector for the primal problem is $\bar{x} = (1, 2)$, yielding a value of $1 + 2 = 3$. The minimizer vector for the dual problem is $\bar{\lambda} = (\frac{2}{5}, \frac{1}{5})$, yielding a value of $4(\frac{2}{5}) + 7(\frac{1}{5}) = \frac{15}{5} = 3$.

# M.N Dynamic Programming

Dynamic programming is a technique for the study of maximization problems defined over sequences that extend to an infinite horizon. We consider here only a very particular and simple case of what is a very general theory [an extensive review is contained in Stokey and Lucas with Prescott (1989)].

Suppose that $A \subset \mathbb{R}^N$ is a nonempty, compact, set.[32] Let $u: A \times A \to \mathbb{R}$ be a continuous function and let $\delta \in (0, 1)$. Given a vector $z \in A$ (interpreted as the initial condition of the variables $\{x_t\}_{t=0}^{\infty}$), the maximization problem we are now interested in is

$$\begin{aligned} \underset{\{x_t\}_{t=0}^{\infty}}{\text{Max}} \quad & \sum_{t=0}^{\infty} \delta^t u(x_t, x_{t+1}) \\ \text{s.t.} \quad & x_t \in A \text{ for every } t, \\ & x_0 = z. \end{aligned} \tag{M.N.1}$$

It is not mathematically difficult to verify that a maximizer sequence exists for problem (M.N.1) and that, therefore, there is a maximum value $v(z)$. The function $v: A \to \mathbb{R}$ is called the *value function* of problem (M.N.1). As with $u(\cdot, \cdot)$ itself, the value function is continuous. If, in addition, $A$ is convex and $u(\cdot, \cdot)$ is concave, then $v(\cdot)$ is also concave.

31. For linear programming problems the constraint qualification is not required. Put another way, the linearity of the constraints is a sufficient form of constraint qualification.

32. The compactness assumption cannot be dispensed with entirely, but it can be much weakened.

It is fairly clear that for every $z \in A$ the value function satisfies the so-called *Bellman equation* (or the *Bellman optimality principle*):

$$v(z) = \underset{z' \in A}{\text{Max}} \quad u(z, z') + \delta v(z').$$

It is perhaps more surprising that, as shown in Theorem M.N.1, the value function is the *only* function that satisfies this equation.

**Theorem M.N.1:** Suppose that $f: A \to \mathbb{R}$ is a continuous function such that for every $z \in A$ the Bellman equation is satisfied; that is,

$$f(z) = \underset{z' \in A}{\text{Max}} \quad u(z, z') + \delta f(z') \tag{M.N.2}$$

for all $z \in A$. Then the function $f(\cdot)$ coincides with $v(\cdot)$; that is, $f(z) = v(z)$ for every $z \in A$.

**Proof:** Successively applying (M.N.2) we have that, for every $T$,

$$\begin{aligned} f(z) = \underset{\{x_t\}_{t=0}^{T}}{\text{Max}} \quad & \sum_{t=0}^{T-1} \delta^t u(x_t, x_{t+1}) + \delta^T f(x_T) \\ \text{s.t. } & x_t \in A \text{ for all } t \leq T, \\ & x_0 = z. \end{aligned}$$

But as $T \to \infty$, the term $\delta^T f(\cdot)$ makes an increasingly negligible contribution to the sum. We conclude therefore that $f(z) = v(z)$. ■

Theorem M.N.1 suggests a procedure for the computation of the value function. Suppose that for $r = 0$ we start with an arbitrary continuous function $f_0: A \to \mathbb{R}$. Think of $f_0(z')$ as a trial "evaluation" function giving a tentative evaluation of the value of starting with $z' \in A$. Then we can generate a new tentative evaluation function $f_1(\cdot)$ by letting, for every $z \in A$,

$$f_1(z) = \underset{z' \in A}{\text{Max}} \quad u(z, z') + \delta f_0(z').$$

If $f_1(\cdot) = f_0(\cdot)$, then $f_0(\cdot)$ satisfies the Bellman equation and Theorem M.N.1 tells us that, in fact, $f_0(\cdot) = v(\cdot)$. If $f_1(\cdot) \neq f_0(\cdot)$, then $f_0(\cdot)$ was not correct. We could then try again, starting with the new tentative $f_1(\cdot)$. This will give us a function $f_2(\cdot)$, and so on for an entire sequence of functions $\{f_r(\cdot)\}_{r=0}^{\infty}$. Does this take us anywhere? The answer is that it does: For every $z \in A$, we have $f_r(z) \to v(z)$ as $r \to \infty$. That is, as $r$ increases, we approach the correct evaluation of $z$.

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Suppose that the sequence $\{\bar{x}_t\}_{t=0}^{\infty}$ is a sequence (or a *trajectory*) that solves the maximization problem (M.N.1). A fortiori, for every $t \geq 1$, the decisions taken at $t$ must be optimal. Examining the sum in (M.N.1), we see that $\bar{x}_t$ must solve

$$\underset{x_t \in A}{\text{Max}} \quad u(\bar{x}_{t-1}, x_t) + \delta u(x_t, \bar{x}_{t+1}). \tag{M.N.3}$$

Assuming that $\bar{x}_t$ is in the interior of $A$, (M.N.3) implies that

$$\frac{\partial u(\bar{x}_{t-1}, \bar{x}_t)}{\partial x_{N+n}} + \delta \frac{\partial u(\bar{x}_t, \bar{x}_{t+1})}{\partial x_n} = 0 \tag{M.N.4}$$

for every $n = 1, \ldots, N$.[33] The necessary conditions captured by (M.N.4) are called the *Euler equations* of problem (M.N.1).

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33. Note that the function $u(\cdot, \cdot)$ has $2N$ arguments, the $N$ variables of the initial period and the $N$ variables of the subsequent period. In condition (M.N.4), the variable $x_n$ is the $n$th component of the initial period, and the variable $x_{N+n}$ is the $n$th component of the subsequent period.

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